

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE
SUBJECT: Data Visualization with Power BI and Tableau
Practical 9: Cleaning and Structuring Data for Analysis

AIM: Cleaning and Structuring Data for Analysis

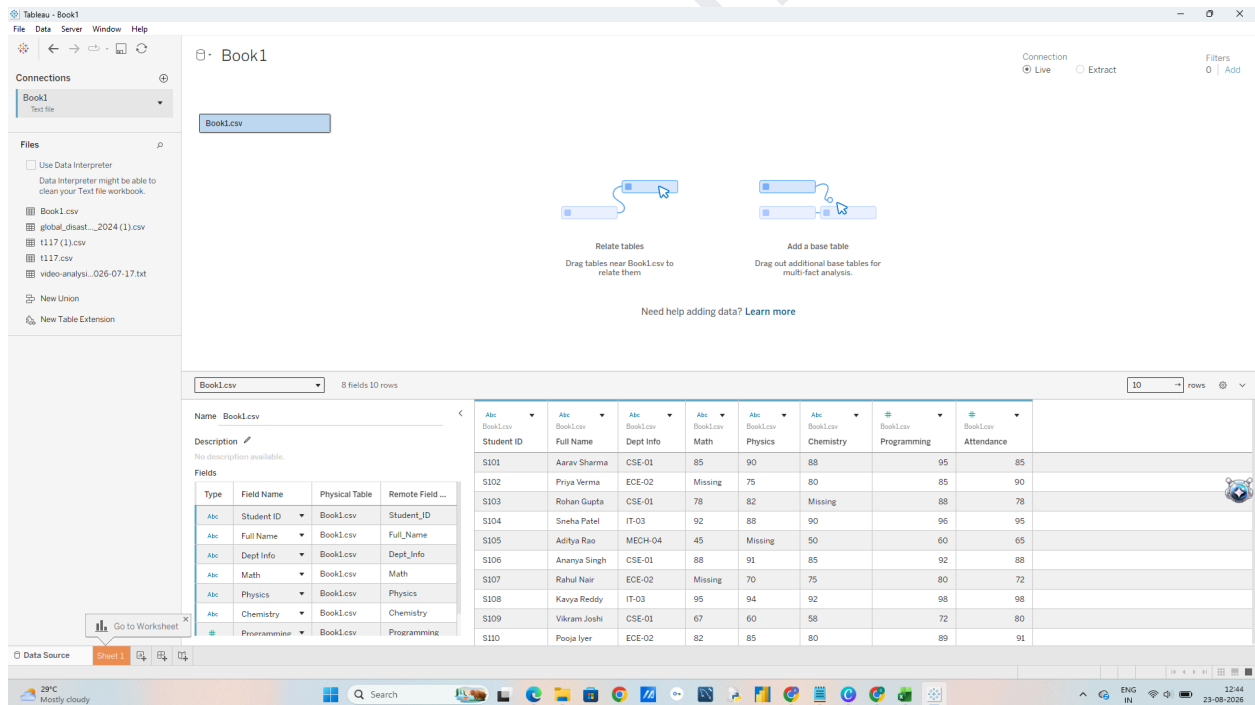
A: Identify and Handle Missing or Inconsistent Data Objective: Detect and resolve data quality issues including missing records (Null values) and data type mismatches (numerical values formatted as text strings with percentage signs).

Tableau treats fields containing non-numeric characters (like % or extra whitespace) as Text (Abc), which disables numerical operations such as calculating averages, standard deviations, or plotting continuous measures.

Missing exam entries (Null) cause dropped rows or incorrect aggregations unless properly imputed or handled.

Step 1: Load the Dataset

1. Open Tableau Desktop.
2. Under the Connect pane on the left, click Text file.
3. Select and open Sem1_Scores.csv.
4. In the Data Source preview table: o Observe missing entries (Null) in Math, Physics, and Chemistry



Name : adarsh shukla

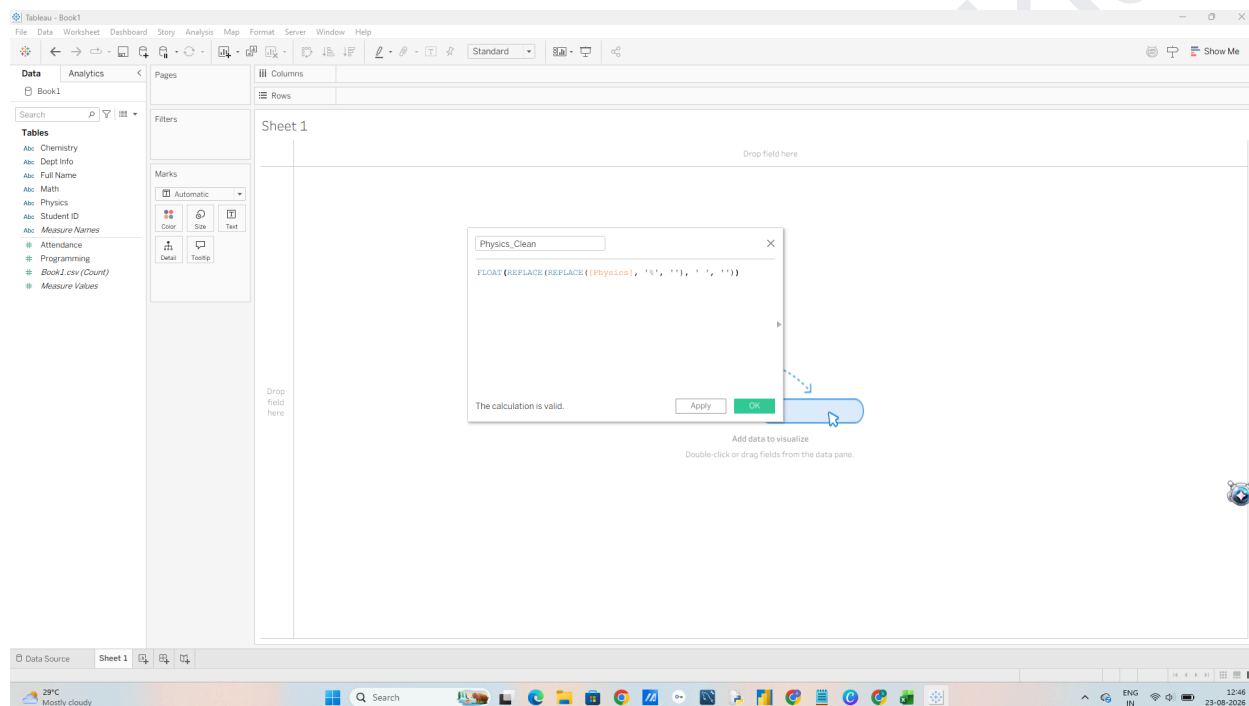
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Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).
2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....
3. Create the cleaned measure for Physics: o Field Name: Physics_Clean o Tableau Formula: Code snippet `FLOAT(REPLACE(STR([Physics]), '%', ''))` o Click OK.



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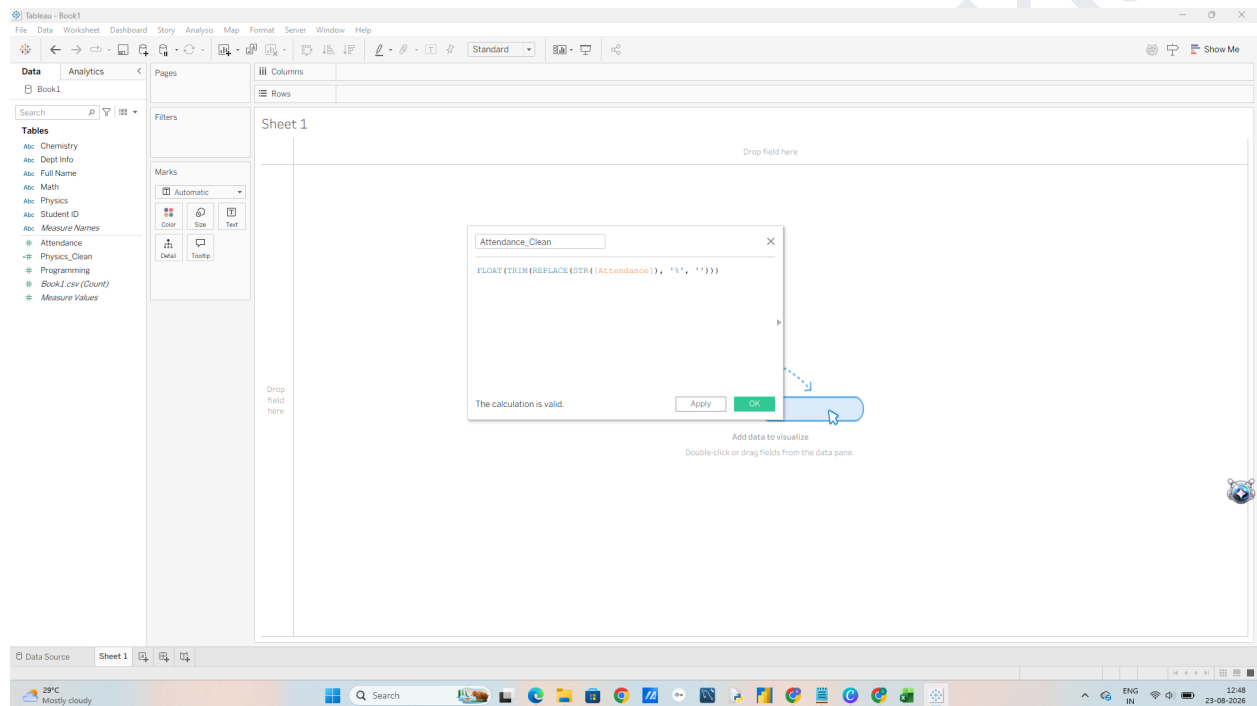
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4. Create the cleaned measure for Attendance: o Right-click the Data Pane --> Create Calculated Field....

o Field Name: Attendance_Clean o Tableau Formula: Code snippet

`FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

o Click OK.



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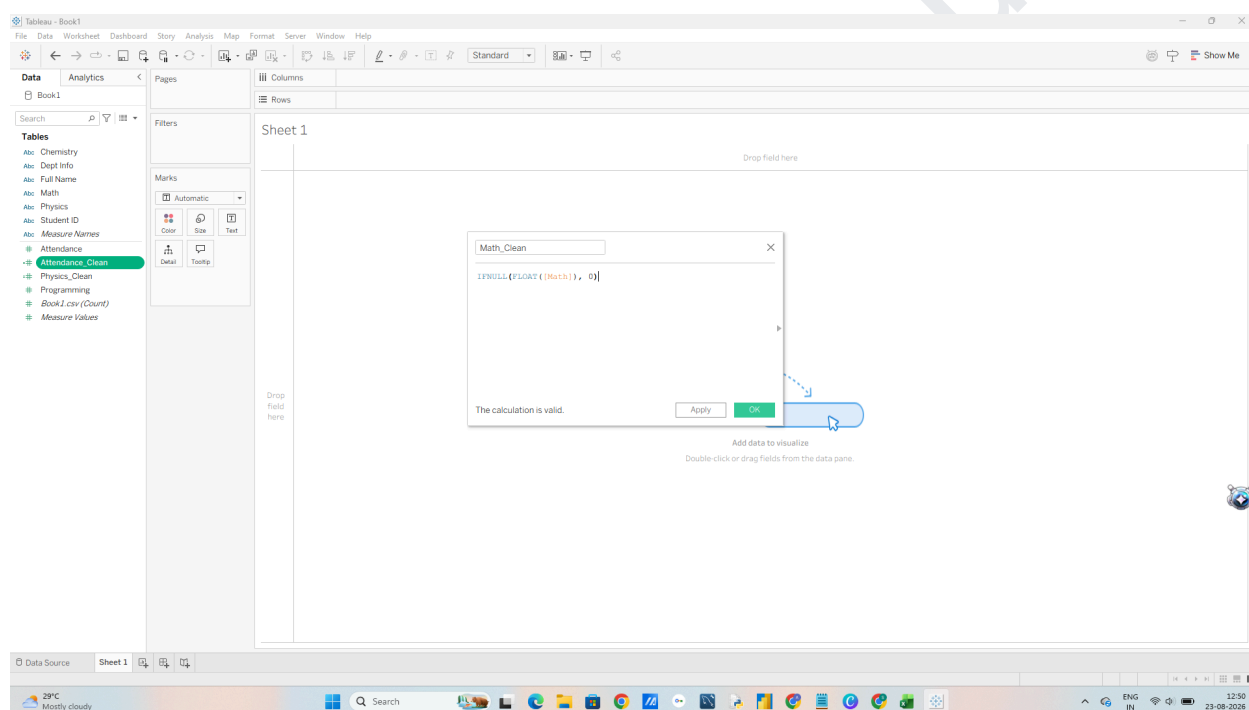
Step 3: Handle Missing (Null) Numeric Values via Imputation

1. Handle missing entries in Math:

o Right-click the Data Pane --> Create Calculated Field....

o Field Name: Math_Clean o Tableau Formula: Code snippet IFNULL([Math], 0)

o Click OK.



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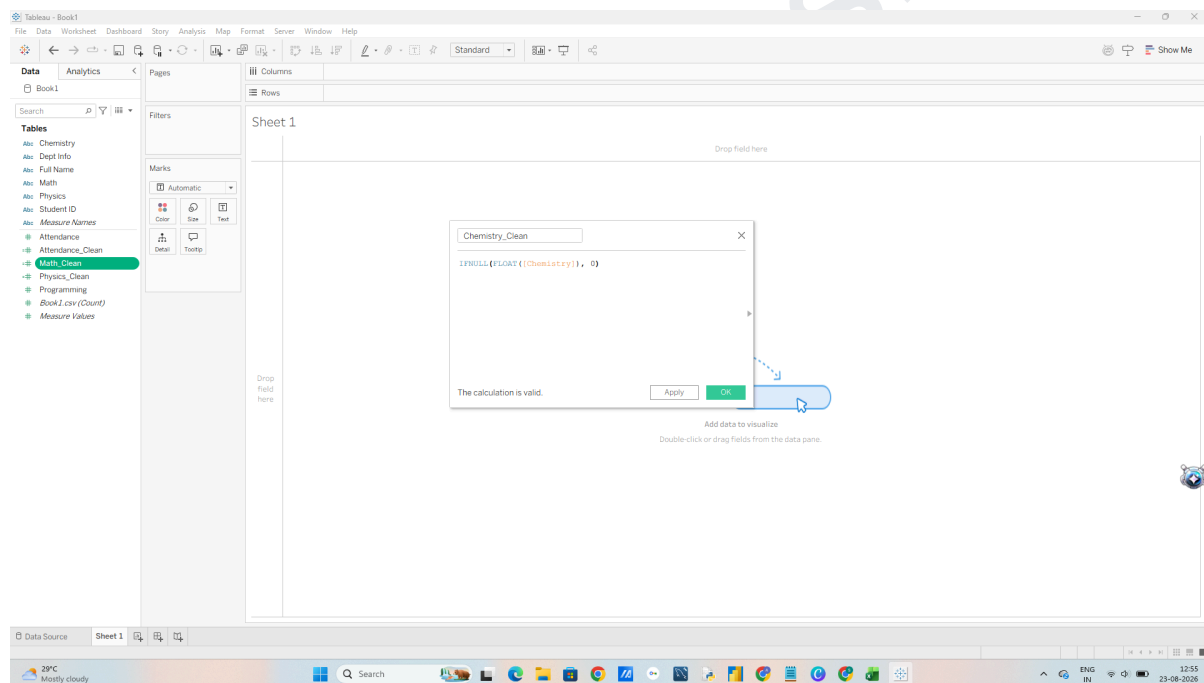
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2. Handle missing entries in Chemistry:

o Right-click the Data Pane --> Create Calculated Field....

o Field Name: Chemistry_Clean o Tableau Formula: Code snippet IFNULL([Chemistry], 0)

o Click OK.



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Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

1. Student S105 contains a Null value in Physics. Combine string stripping with null handling:

- o Edit Physics_Clean (right-click Physics_Clean --> Edit...):

- o Updated Tableau Formula: Code snippet IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0

- o Click OK.

B: Convert Wide Data into Tall Format Objective: Transform the horizontal subject columns (Math, Physics, Chemistry, Programming) into a normalized, vertical key-value pair: Subject (Dimension) and Score(Measure). In wide format, comparing performance across subjects requires manually adding multiple distinct measures to shelves.

Unpivoting to tall (tidy) format collapses four separate columns into a single categorical dimension (Subject), enabling dynamic colour coding, subject-wise side-by-side bar charts, and streamlined slice-and-dice filtering in Tableau.

Step 1: Open the Data Source Grid

1. In Tableau, navigate to the Data Source tab at the bottom-left corner.

2. Locate the four horizontal subject score columns:

- o Math

- o Physics

- o Chemistry

- o Programming

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The screenshot shows the Tableau Desktop interface with a data source named 'Book1' (Text file) connected. The data is displayed in a table with 13 fields and 10 rows. The columns are: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, Attendance, Physics_Clean, Attendance_Clean, and Math_... (partially visible). The 'Math' column is highlighted in blue. The 'Attendance' column is also highlighted in blue. The 'Physics_Clean' column is highlighted in blue. The 'Attendance_Clean' column is highlighted in blue. The 'Math_...' column is highlighted in blue. The 'Math' column is highlighted in blue. The 'Attendance' column is highlighted in blue. The 'Physics_Clean' column is highlighted in blue. The 'Attendance_Clean' column is highlighted in blue. The 'Math_...' column is highlighted in blue.

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean	Math_...
S101	Aarav Sharma	CSE-01	85	90	88	95	85	90.0000	85.0000	
S102	Priya Verma	ECE-02	Missing	75	80	85	90	75.0000	90.0000	
S103	Rohan Gupta	CSE-01	78	82	Missing	88	78	82.0000	78.0000	
S104	Sneha Patel	IT-03	92	88	90	96	95	88.0000	95.0000	
S105	Aditya Rao	MECH-04	45	Missing	50	60	65	null	65.0000	
S106	Ananya Singh	CSE-01	88	91	85	92	88	91.0000	88.0000	
S107	Rahul Nair	ECE-02	Missing	70	75	80	72	70.0000	72.0000	
S108	Kavya Reddy	IT-03	95	94	92	98	98	94.0000	98.0000	
S109	Vikram Joshi	CSE-01	67	60	58	72	80	60.0000	80.0000	
S110	Pooja Iyer	ECE-02	82	85	80	89	91	85.0000	91.0000	

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue

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Tableau - Book1

File Data Server Window Help

Connections

Book1

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Book1.csv

global_distant...2024 (1).csv

t117 (1).csv

t117.csv

video-analysi...026-07-17.txt

New Union

New Table Extension

Book1.csv

13 fields 10 rows

Name Book1.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field ...
+	Programming	Book1.csv	Programming
+	Attendance	Book1.csv	Attendance
+	Physics_Clean	Calculation	Calculation_137...
+	Attendance_C	Calculation	Calculation_137...
+	Math_Clean	Calculation	Calculation_137...
+	Chemistry_Cl	Calculation	Calculation_137...
+	Physics_Clean	Calculation	Calculation_137...

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean	Math_Clean
S101	Aarav Sharma	CSE-01	85	90	88	95	85	90.0000	85.0000	
S102	Priya Verma	ECE-02	Missing	75	80	85	90	75.0000	90.0000	
S103	Rohan Gupta	CSE-01	78	82	Missing	88	78	82.0000	78.0000	
S104	Sneha Patel	IT-03	92	88	90	96	95	88.0000	95.0000	
S105	Aditya Rao	MECH-04	45	Missing	50	60	65	null	65.0000	
S106	Ananya Singh	CSE-01	88	91	85	92	88	91.0000	88.0000	
S107	Rahul Nair	ECE-02	Missing	70	75	80	72	70.0000	72.0000	
S108	Kanya Reddy	IT-03	95	94	92	98	98	94.0000	98.0000	
S109	Vikram Joshi	CSE-01	67	60	58	72	80	60.0000	80.0000	
S110	Pooja Iyer	ECE-02	82	85	80	89	91	85.0000	91.0000	

30°C Mostly cloudy

Search

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3. Right-click on any of the highlighted column headers.

4. Select Pivot.

Tableau - Book1

File Data Server Window Help

Connections

Book1

Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

Book1.csv

global_distant...2024 (1).csv

t117 (1).csv

t117.csv

video-analysi...026-07-17.txt

New Union

New Table Extension

Book1.csv

13 fields 10 rows

Name Book1.csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Remote Field ...
+	Student ID	Book1.csv	Student ID
+	Full Name	Book1.csv	Full Name
+	Dept Info	Book1.csv	Dept Info
+	Math	Book1.csv	Math
+	Physics	Book1.csv	Physics
+	Chemistry	Book1.csv	Chemistry
+	Programming	Book1.csv	Programming
+	Attendance	Book1.csv	Attendance
+	Physics_Clean	Calculation	Calculation_137...
+	Attendance_Clean	Calculation	Calculation_137...
+	Math_Clean	Calculation	Calculation_137...
+	Chemistry_Clean	Calculation	Calculation_137...
+	Physics_Clean	Calculation	Calculation_137...

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Attendance_Clean	Math_Clean	Chemistry_Clean	Physics_Clean
S101	Aarav Sharma	CSE-01	85	90	88	95	85	90.0000	85.0000	85.0000	88.0000	90.0000
S102	Priya Verma	ECE-02	Missing	75	80	85	90	75.0000	90.0000	0.0000	80.0000	75.0000
S103	Rohan Gupta	CSE-01	78	82	Missing	88	78	82.0000	78.0000	78.0000	0.0000	82.0000
S104	Sneha Patel	IT-03	92	88	90	96	95	88.0000	95.0000	92.0000	90.0000	88.0000
S105	Aditya Rao	MECH-04	45	Missing	50	60	65	null	65.0000	45.0000	50.0000	0.0000
S106	Ananya Singh	CSE-01	88	91	85	92	88	91.0000	88.0000	88.0000	85.0000	91.0000
S107	Rahul Nair	ECE-02	Missing	70	75	80	72	70.0000	72.0000	0.0000	75.0000	70.0000
S108	Kanya Reddy	IT-03	95	94	92	98	98	94.0000	98.0000	95.0000	92.0000	94.0000
S109	Vikram Joshi	CSE-01	67	60	58	72	80	60.0000	80.0000	67.0000	58.0000	60.0000
S110	Pooja Iyer	ECE-02	82	85	80	89	91	85.0000	91.0000	82.0000	80.0000	85.0000

30°C Mostly cloudy

Search

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Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

1. Double-click the header named Pivot Field Names and type:

Subject2. Double-click the header named Pivot Field Values and type: Score

The screenshot displays the Tableau interface for a workbook named 'Book1'. The data source is 'Book1.csv', which contains 11 fields and 40 rows. The fields are categorized as follows:

Type	Field Name	Physical Table	Remote Field ...
ABC	Student ID	Book1.csv	Student_ID
ABC	Full Name	Book1.csv	Full_Name
ABC	Dept Info	Book1.csv	Dept_Info
ABC	Subject	Book1.csv	Subject
ABC	Pivot Field No	Pivot	Pivot Field Nam...
ABC	Pivot Field Val	Pivot	Pivot Field Values
ABC	Physics_Cle 1	Calculation	Calculation_137...

The pivot table below shows the following columns:

Book1.csv	Book1.csv	Book1.csv	Book1.csv	Calculation	Calculation	Calculation	Calculation	Calculation
Dept Info	Subject	Pivot Field Names	Pivot Field Values	Physics_Clean	Attendance_Clean	Math_Clean	Chemistry_Clean	Physics_Clean
CSE-01	85	Chemistry	88	null	85.0000	null	null	null
CSE-01	85	Math	85	null	85.0000	null	null	null
CSE-01	85	Physics	90	null	85.0000	null	null	null
CSE-01	85	Programming	95	null	85.0000	null	null	null
ECE-02	90	Chemistry	80	null	90.0000	null	null	null
ECE-02	90	Math	Missing	null	90.0000	null	null	null
ECE-02	90	Physics	75	null	90.0000	null	null	null
ECE-02	90	Programming	85	null	90.0000	null	null	null
CSE-01	78	Chemistry	Missing	null	78.0000	null	null	null
CSE-01	78	Math	78	null	78.0000	null	null	null

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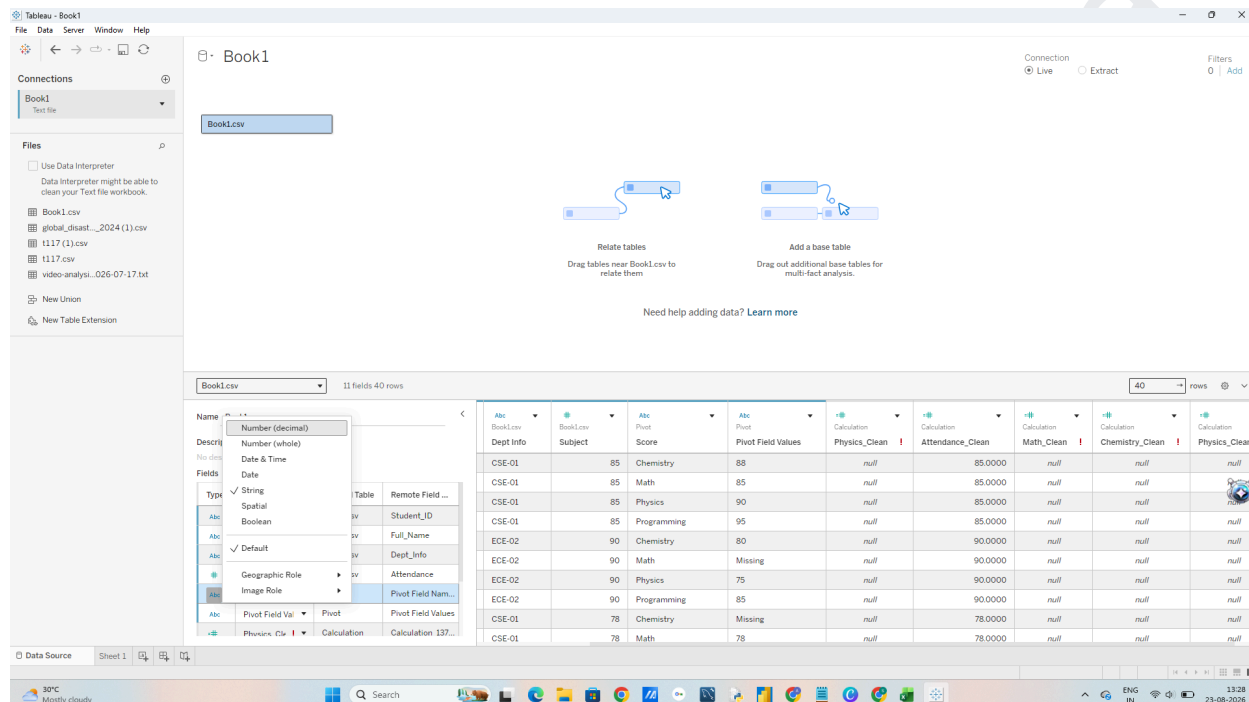
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Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure:

1. Click the Abc icon directly above the Score column header.
2. Select Number (decimal) from the dropdown list



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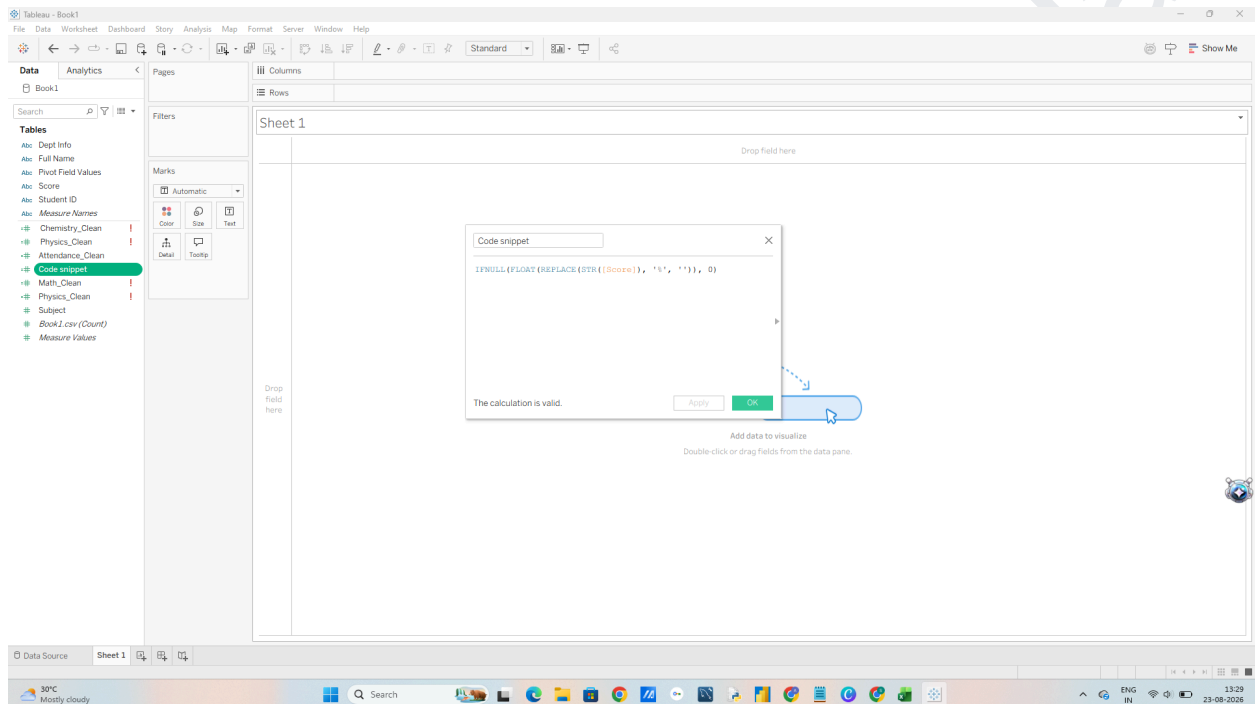
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3. Alternatively, create a calculated field inside

Sheet 1 named Clean_Score:

Code snippet IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)



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C Perform Union of Multiple Files Objective: Stack Sem1_Scores.csv and Sem2_Scores.csv vertically into a single unified table. In academic reporting, performance data is recorded semester by semester in distinct files with identical schemas. Performing a Union consolidates multiple periodic files into a single long dataset, allowing you to track semester-over-semester score trends, compare

Semester 1 versus Semester

2 performance for each student, and compute overall year-long averages.

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Book1' as a Text File. Below it, the 'Files' pane lists several CSV files: 'Book1.csv', 'global_disast..._2024 (1).csv', 't117 (1).csv', 't117.csv', and 'video-analys..._026-07-17.txt'. The main canvas area shows 'Book1' with a right-click context menu open, offering options like 'Open...', 'Rename', 'Remove', 'Field names are in first row', 'Generate field names automatically', 'Text File Properties...', and 'Convert to Union...'. Below the canvas, a data preview table is visible, showing columns for 'Book1.csv', 'Dept Info', 'Subject', 'Score', 'Pivot Field Values', and several calculated fields for 'Physics_Clean', 'Attendance_Clean', 'Math_Clean', 'Chemistry_Clean', and 'Physics_Clean'. The table contains data for various subjects and scores, with some rows showing null values.

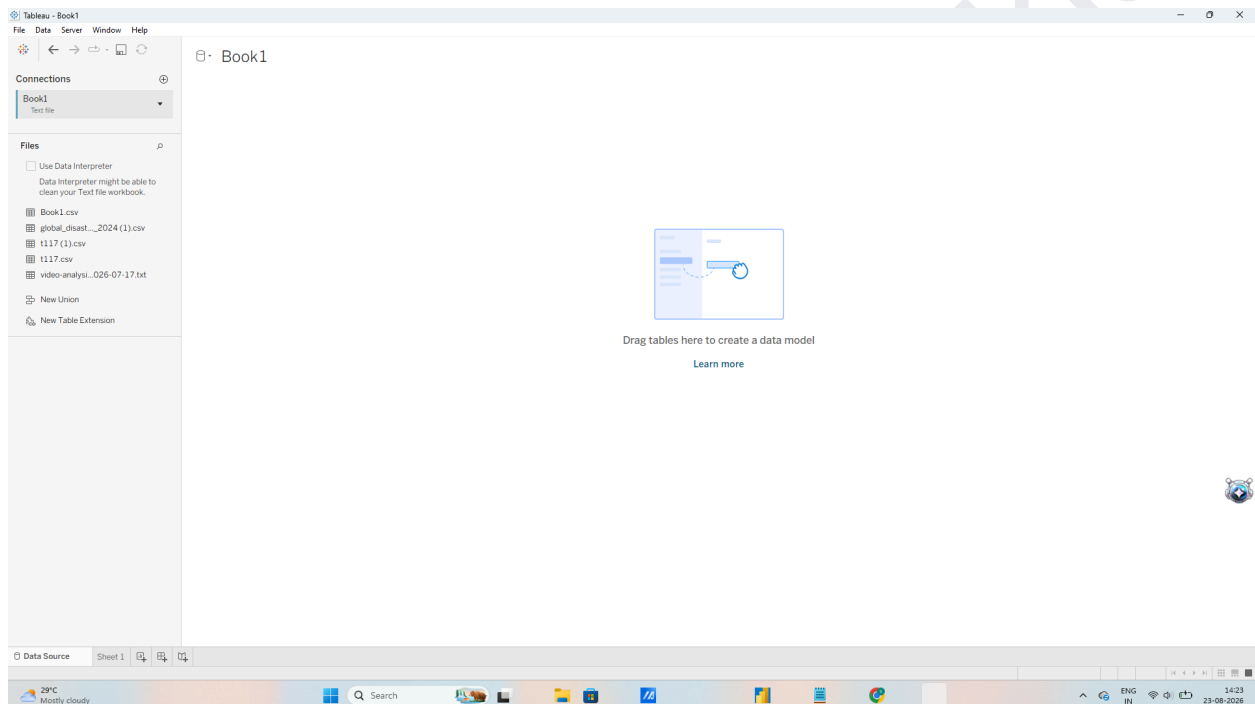
Book1.csv	Dept Info	Subject	Score	Pivot Field Values	Physics_Clean	Attendance_Clean	Math_Clean	Chemistry_Clean	Physics_Clean
CSE-01	85	Chemistry	88		null	85.0000	null	null	null
CSE-01	85	Math	85		null	85.0000	null	null	null
CSE-01	85	Physics	90		null	85.0000	null	null	null
CSE-01	85	Programming	95		null	85.0000	null	null	null
ECE-02	90	Chemistry	80		null	90.0000	null	null	null
ECE-02	90	Math	Missing		null	90.0000	null	null	null
ECE-02	90	Physics	75		null	90.0000	null	null	null
ECE-02	90	Programming	85		null	90.0000	null	null	null
CSE-01	78	Chemistry	Missing		null	78.0000	null	null	null
CSE-01	78	Math	78		null	78.0000	null	null	null

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Step 2: Create a Multi-File Union 1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.



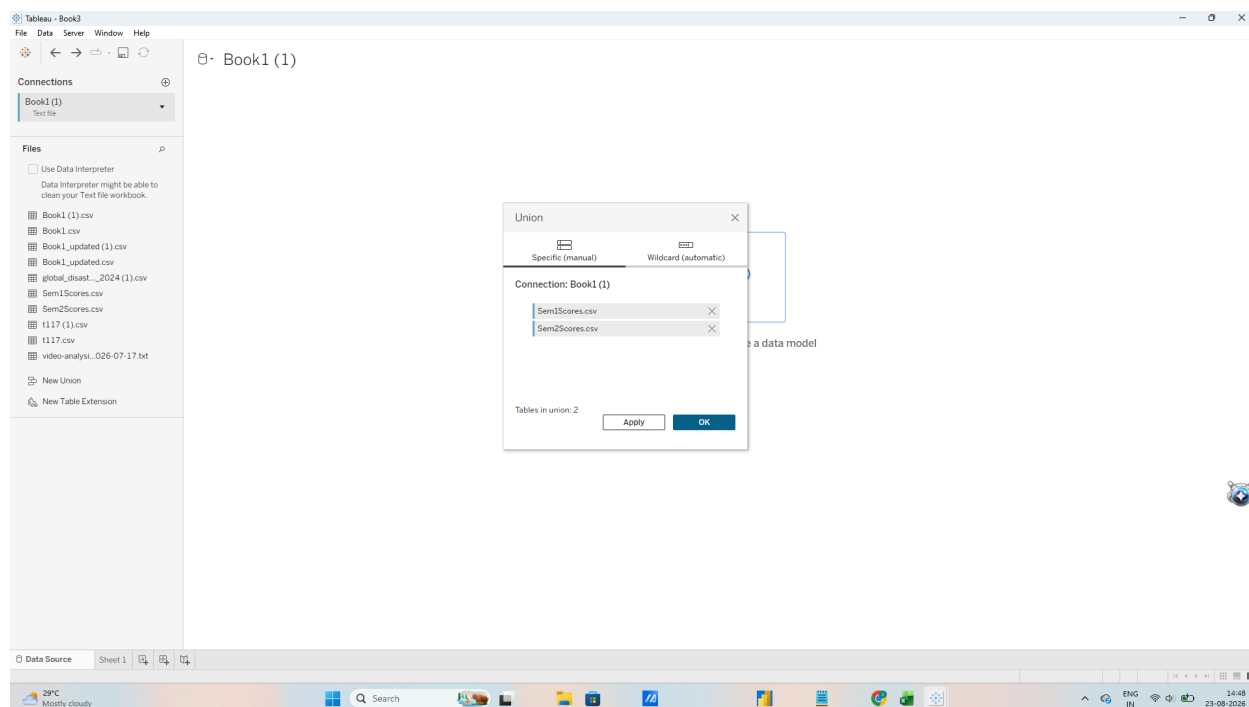
3. In the pop-up Union dialog box:
o Drag Sem1_Scores.csv from the left panel into the
o Drag Sem2_Scores.csv from the left panel into the same box.
o Verify that both filenames are listed in the Union box.
4. Click Apply, then click OK.

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5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.

Step 3: Verify the Merged Data Grid 1. Review the data grid at the bottom of the screen. 2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).

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The screenshot displays the Tableau Desktop interface. On the left, the 'Connections' pane shows a list of files, including 'Book1 (1)' and various CSV files. The 'Files' pane lists several CSV files, including 'Book1 (1).csv', 'Book1_updated.csv', 'global_data_2024 (1).csv', 'Sem1Scores.csv', 'Sem2Scores.csv', 't117 (1).csv', and 'video-analysis_026-07-17.txt'. The 'New Union' button is visible. The main workspace shows a 'Union' operation with a data grid. The grid has 9 columns: 'Student_ID', 'Full_Name', 'Dept_Info', 'Math', 'Physics', 'Chemistry', 'Programming', 'Attendance', and 'Table Name'. The 'Table Name' column shows 'Sem1_Scores.csv' for the first 10 rows and 'Sem2_Scores.csv' for the remaining 10 rows. The bottom status bar shows '23°C Mostly cloudy' and '14:48 23-08-2026'.

Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S101	Aarav Sharma	CSE-01	85	0.900000	88	95	0.850000	Sem1Scores.csv
S102	Priya Verma	ECE-02	null	0.750000	80	85	0.900000	Sem1Scores.csv
S103	Rohan Gupta	CSE-01	78	0.820000	null	88	0.780000	Sem1Scores.csv
S104	Sneha Patel	IT-03	92	0.880000	90	96	0.950000	Sem1Scores.csv
S105	Aditya Rao	MECH-04	45	null	50	60	0.650000	Sem1Scores.csv
S106	Ananya Singh	CSE-01	88	0.910000	85	92	0.880000	Sem1Scores.csv
S107	Rahul Nair	ECE-02	null	0.700000	75	80	0.720000	Sem1Scores.csv
S108	Kavya Reddy	IT-03	95	0.940000	92	98	0.980000	Sem1Scores.csv
S109	Vikram Joshi	CSE-01	67	0.600000	58	72	0.800000	Sem1Scores.csv
S110	Pooja Iyer	ECE-02	82	0.850000	80	89	0.910000	Sem1Scores.csv

3. Scroll to the far right of the grid to verify the generated system column:
o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

C: Perform Union of Multiple Files Objective: Stack Sem1_Scores.csv and Sem2_Scores.csv vertically into a single unified table. In academic reporting, performance data is recorded semester by semester in distinct files with identical schemas. Performing a Union consolidates

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multiple periodic files into a single long dataset, allowing you to track semester-over-semester score trends, compare

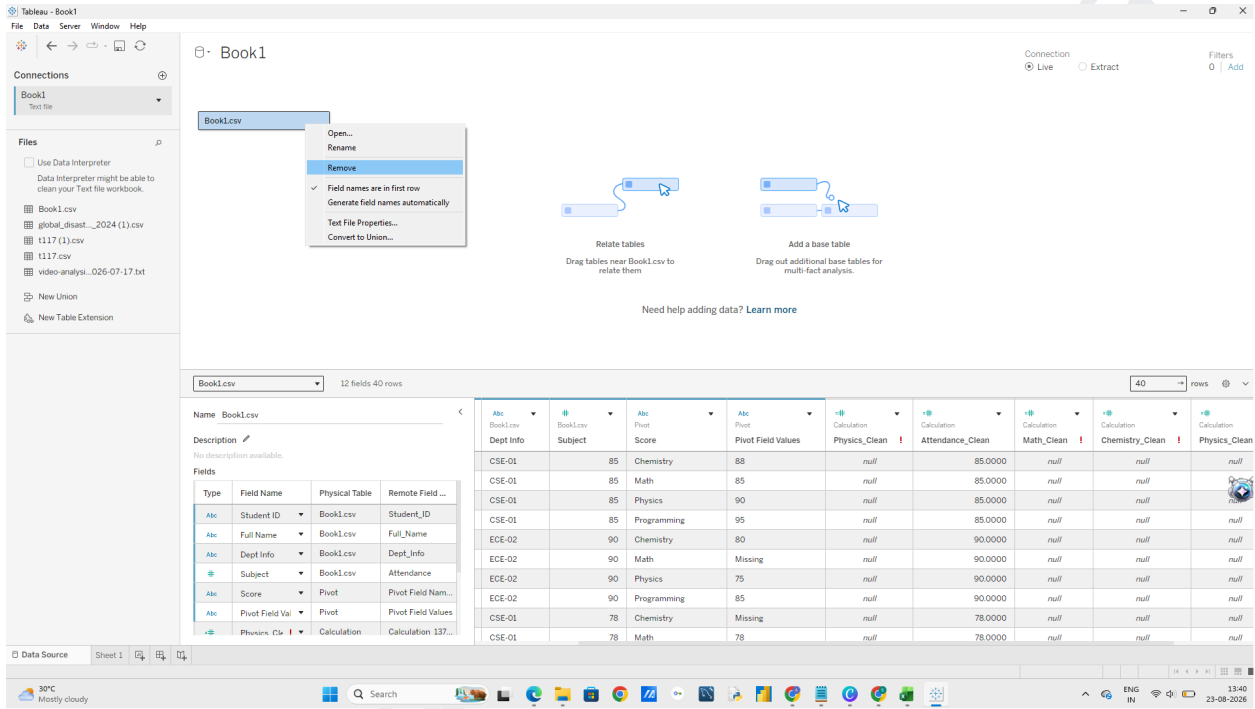
Semester 1 versus Semester

2 performance for each student, and compute overall year-long averages.

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.

2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.



The screenshot shows the Tableau Desktop interface. In the bottom-left pane, the 'Connections' tab is active, showing a list of files including 'Book1.csv'. A context menu is open over 'Book1.csv', with the 'Remove' option selected. The main canvas displays a data preview for 'Book1.csv', showing 12 fields and 40 rows. The fields are categorized into 'Physical Table' and 'Remote Field ...'.

Type	Field Name	Physical Table	Remote Field ...
Abc	Student ID	Book1.csv	Student_ID
Abc	Full Name	Book1.csv	Full_Name
Abc	Dept Info	Book1.csv	Dept_Info
+	Subject	Book1.csv	Attendance
Abc	Score	Pivot	Pivot Field Nam...
Abc	Pivot Field Val	Pivot	Pivot Field Values
+	Physics_Clean	Calculation	Calculation 137...

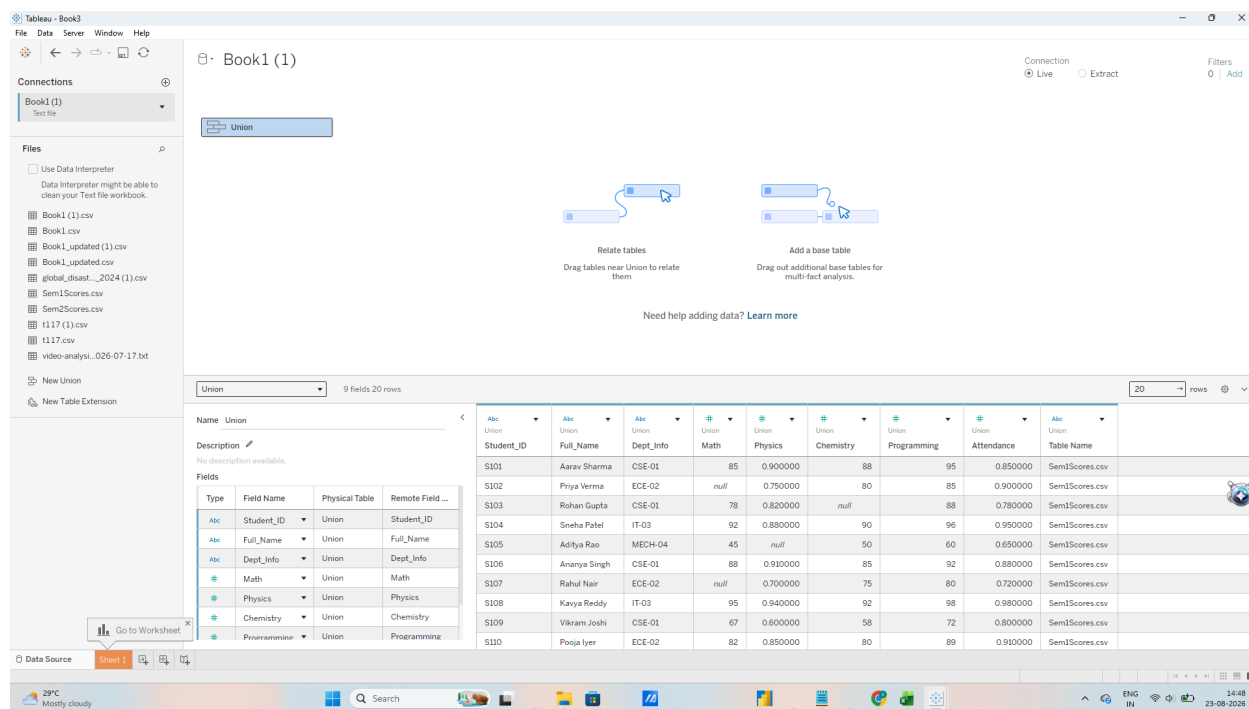
The data preview table shows the following columns: Dept Info, Subject, Score, Pivot Field Values, Physics_Clean, Attendance_Clean, Math_Clean, Chemistry_Clean, and Physics_Clean. The rows represent individual students and their scores across different subjects and semesters.

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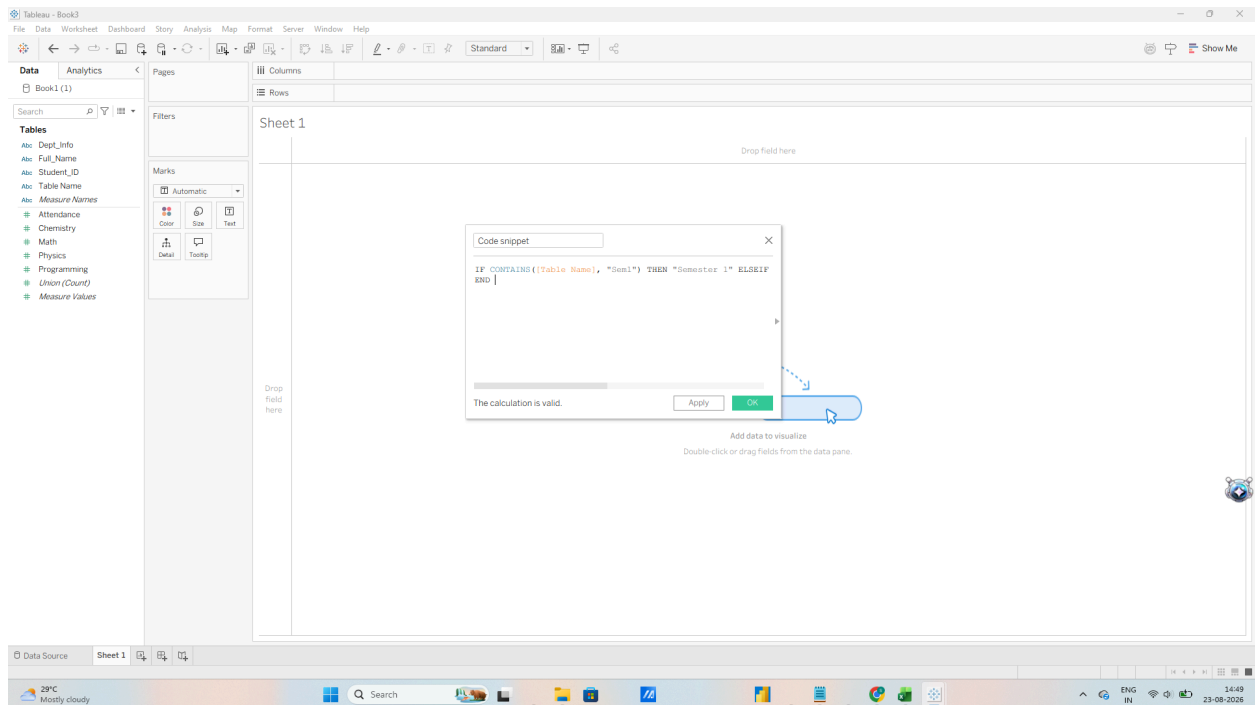
Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
3. Set the Field Name to: Semester
4. Enter the following formula:

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5. Click OK.

D .Execute Cross-Database Joins Objective: Merge your unioned student scores dataset with two independent tables: Department_Master.csv (using Dept_Info = Dept_Code) and Student_Activities.csv(using Student_ID). Student academic scores rarely include institutional metadata (like Department Names or Targets) or extracurricular records.

A Physical Join bridges separate tabular datasets horizontally across common primary/foreign keys, creating an enriched single record for each student.

Step 1: Stay on the Top Canvas Leave your Union table box as it is on the canvas.

Step 2: Add and Link Department_Master.csv 1.

In the left sidebar under Files / Connections, find Department_Master.csv.

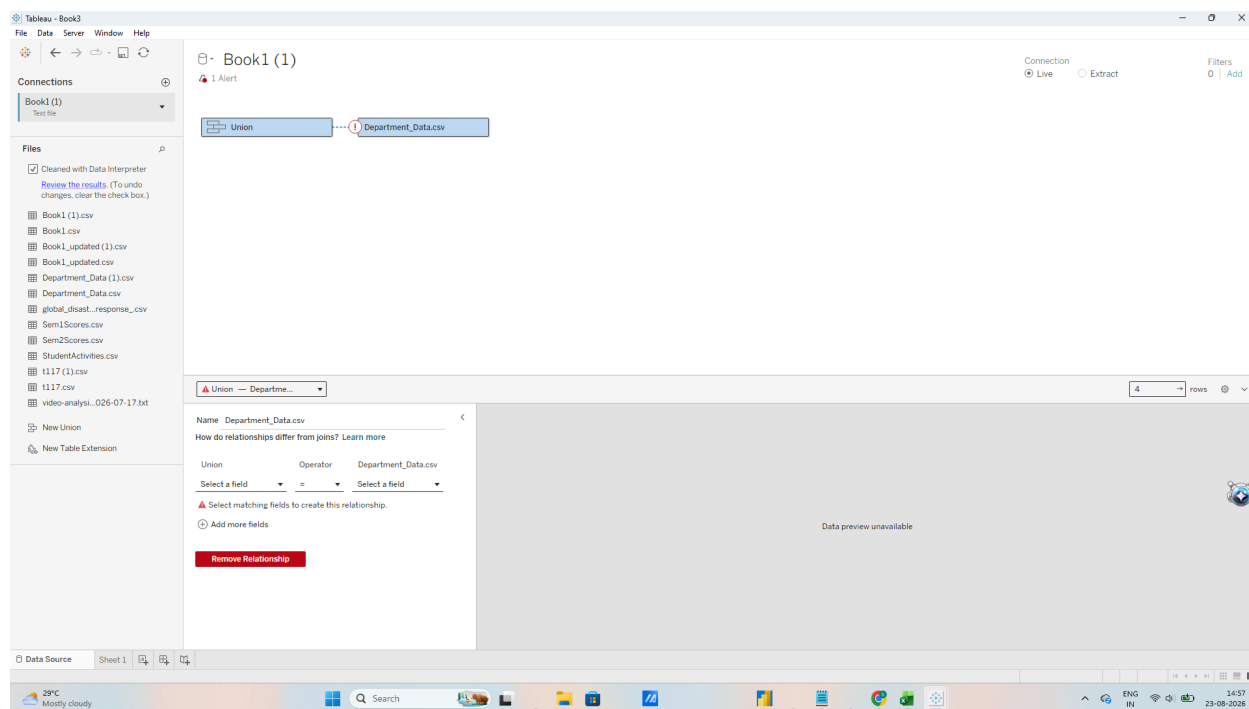
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Unionbox. 3. A flexible line (relationship / join noodle) will connect the two boxes.

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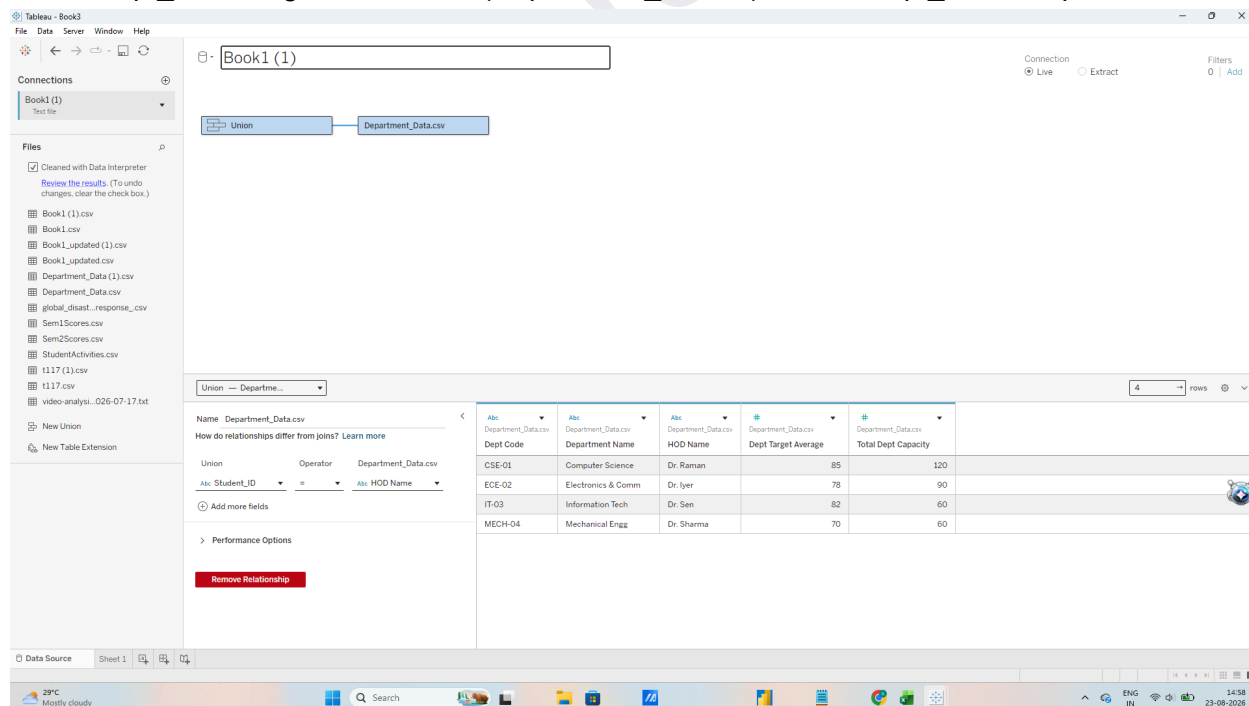
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4. If the relationship dialog opens (or click the relationship line):
o Left Table Field (Union): Select Dept_Info
o Right Table Field (Department_Master): Select Dept_Code
o Operator: =



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Step 3: Add and Link Student_Activities.csv 1. From the left sidebar, drag Student_Activities.csv onto the canvas. 2. Drop it to the right of the Union box.

The screenshot shows the Tableau Desktop interface. In the left sidebar, the 'Connections' pane shows 'Book1 (1)' as the selected connection. The 'Files' pane lists various CSV files, including 'Book1 (1).csv', 'Book1.csv', 'Book1_updated (1).csv', 'Book1_updated.csv', 'Department_Data (1).csv', 'Department_Data.csv', 'global_diseas..._response.csv', 'Sem1Scores.csv', 'Sem2Scores.csv', 'StudentActivities.csv', 't117 (1).csv', 't117.csv', and 'video-analysi..._026-07-17.txt'. The main canvas displays a 'Union' relationship between 'Department_Data.csv' and 'StudentActivities.csv'. The data preview shows the following table:

Student ID (StudentActivi	Club Membership	Projects Completed	Certifications
S101	Coding Club	3	2
S102	Robotics Club	2	1
S103	Coding Club	2	1
S104	Debate Society	4	3
S105	Sports Club	1	0
S106	Coding Club	3	2
S107	Robotics Club	2	1
S108	Debate Society	4	3
S109	Sports Club	1	1
S110	Cultural Club	2	2

3. In the relationship dialog:
- o Left Table Field (Union): Select Student_ID
 - o Right Table Field (Student_Activities): Select Student_ID
 - o Operator: =

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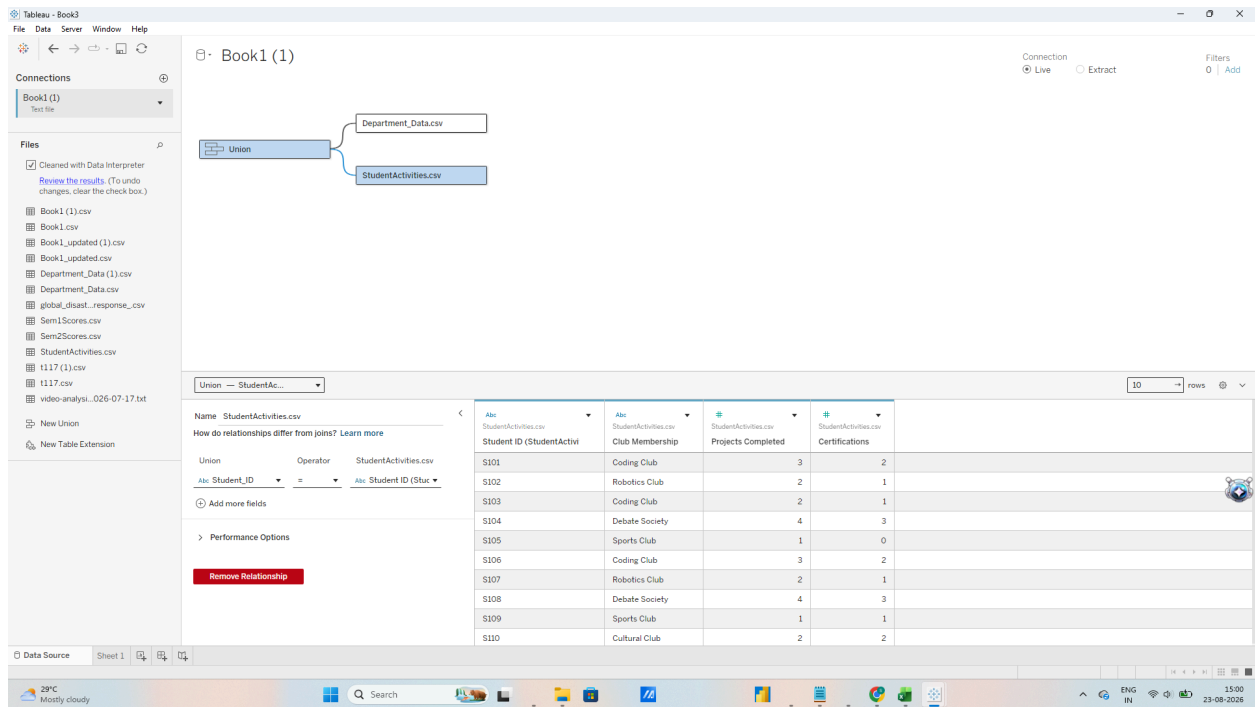
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Step 4: Verify in Sheet 1 1. Click on Sheet 1 at the bottom-left.

2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:

- o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
- o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
- o Student_Activities: Club_Membership, Projects_Completed, Certifications

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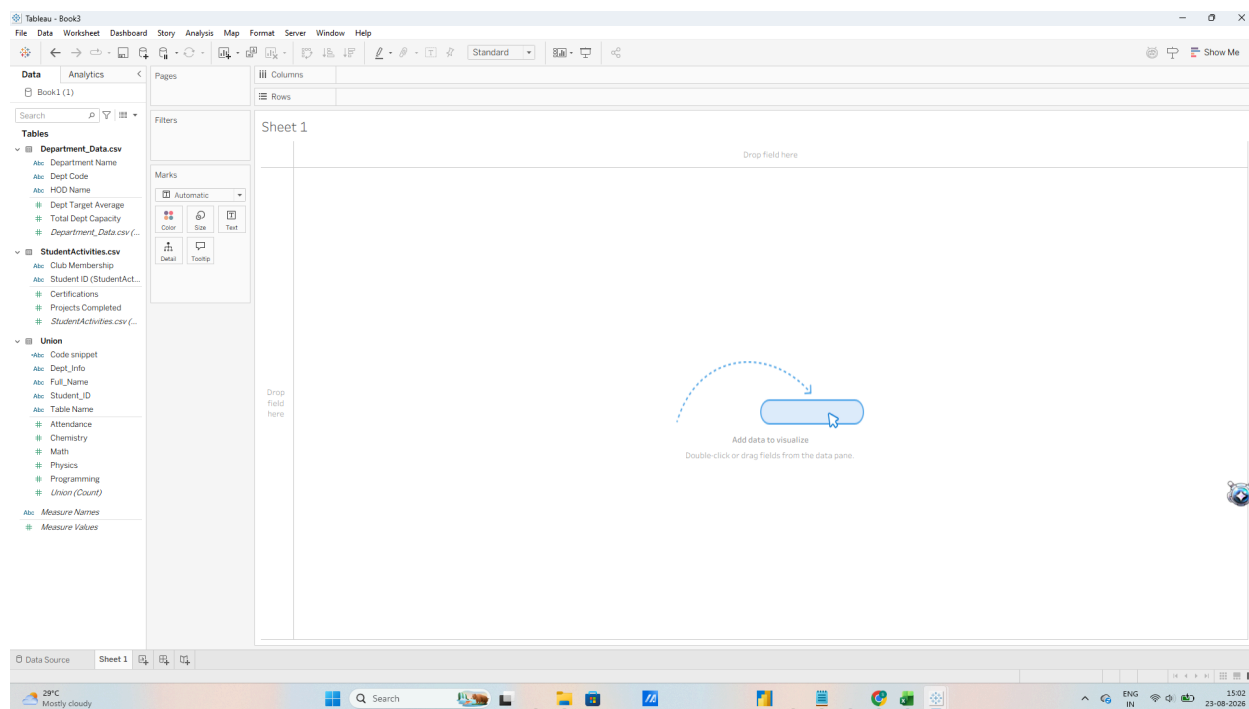
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E: Restructure Data for Better Visualization

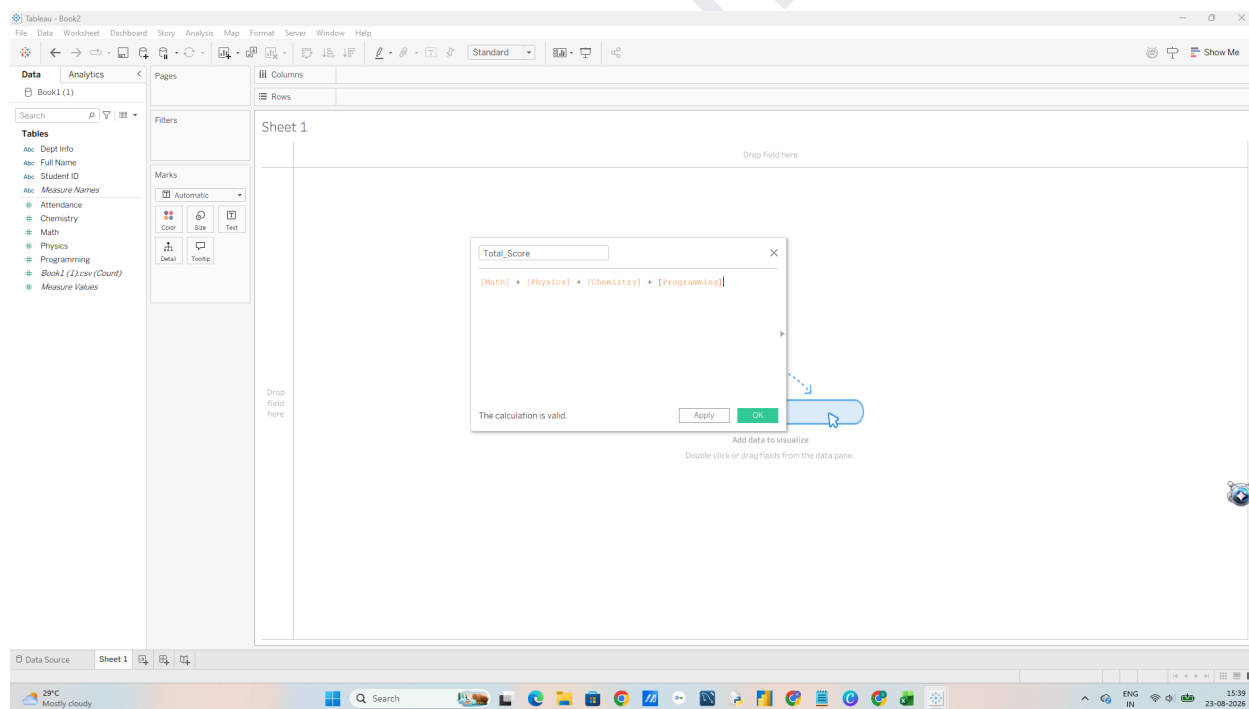
Objective: Create standardized metric fields, calculated performance tiers, and formatted indicators so that raw marks transform into an intuitive student self-performance dashboard.

Raw numerical scores alone do not tell a student whether they are meeting performance criteria or lagging behind.

By restructuring the data into calculated total marks, percentage scores, performance bands (Distinction, First Class, Needs Support), and attendance flags, we prepare the dataset for high-impact visual scorecards, color-coded heatmaps, and target-vs-actual comparison charts.

Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau.
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the student's total marks field:
 - o Field Name: Total_Score
 - o Tableau Formula: Code snippet `[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]`
 - o Click OK.



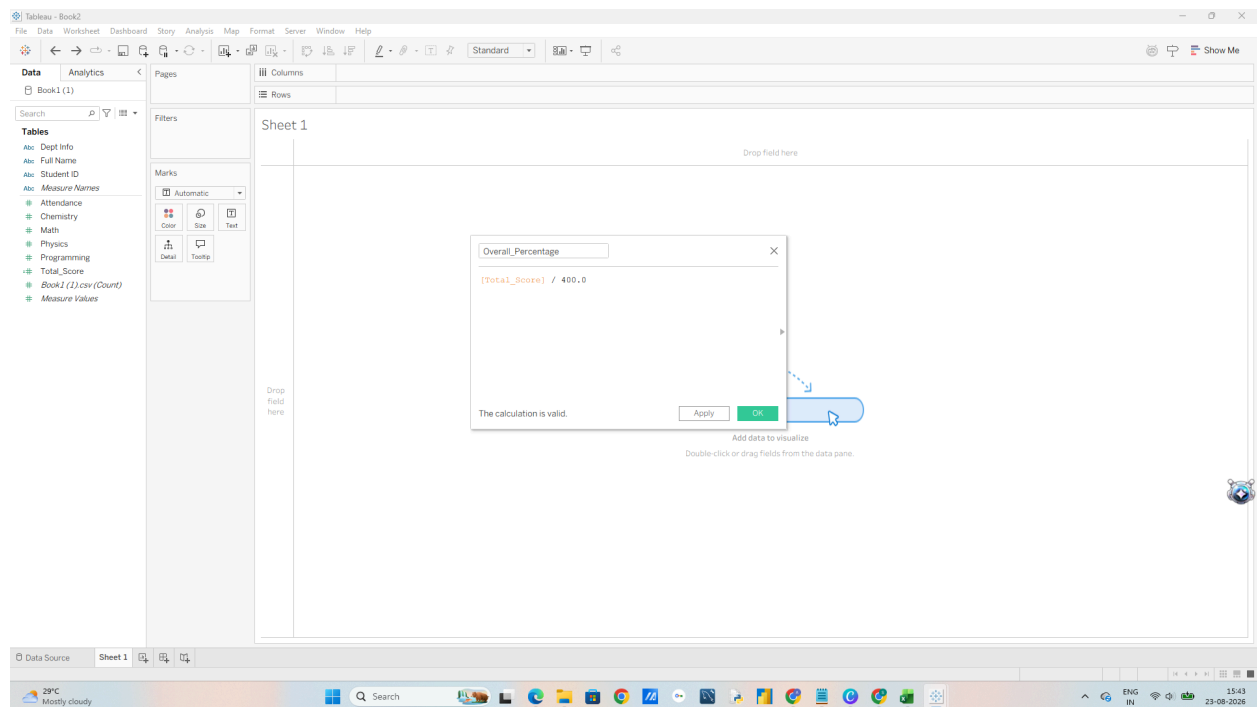
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4. Create the scaled overall percentage field:
- o Right-click in the Data Pane --> select Create Calculated Field....
 - o Field Name: Overall_Percentage o Tableau Formula: Code snippet $[Total_Score] / 400.0$
 - o Click OK.



5. Set the default percentage display formatting:
- o In the Data Pane, right-click Overall_Percentage -->Default Properties -->Number Format....

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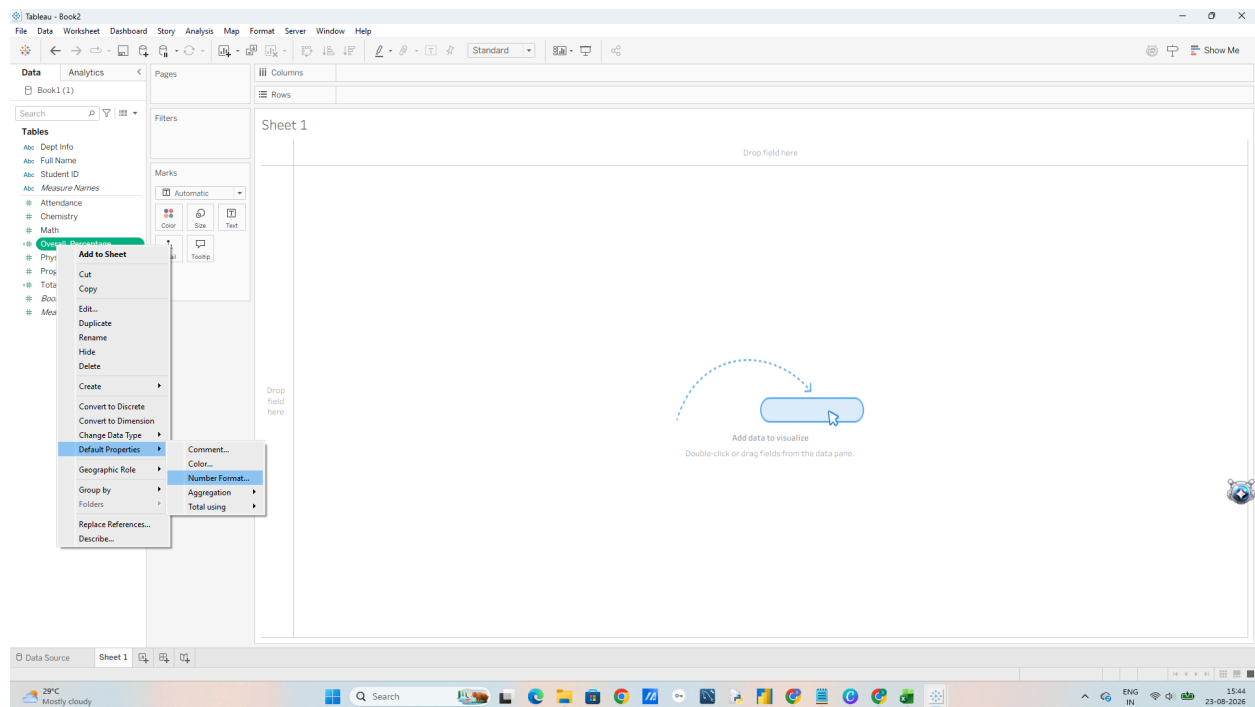
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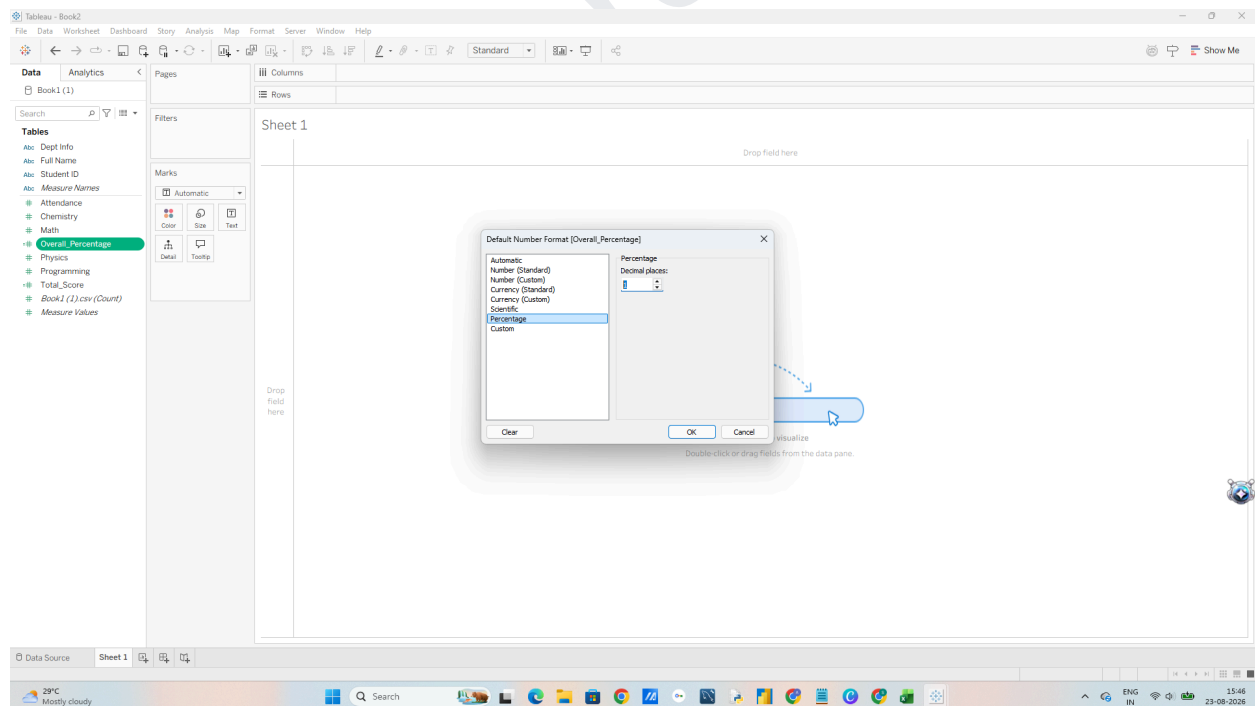
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o Select Percentage and set decimal places to 1 (e.g., 85.5%).

o Click OK.



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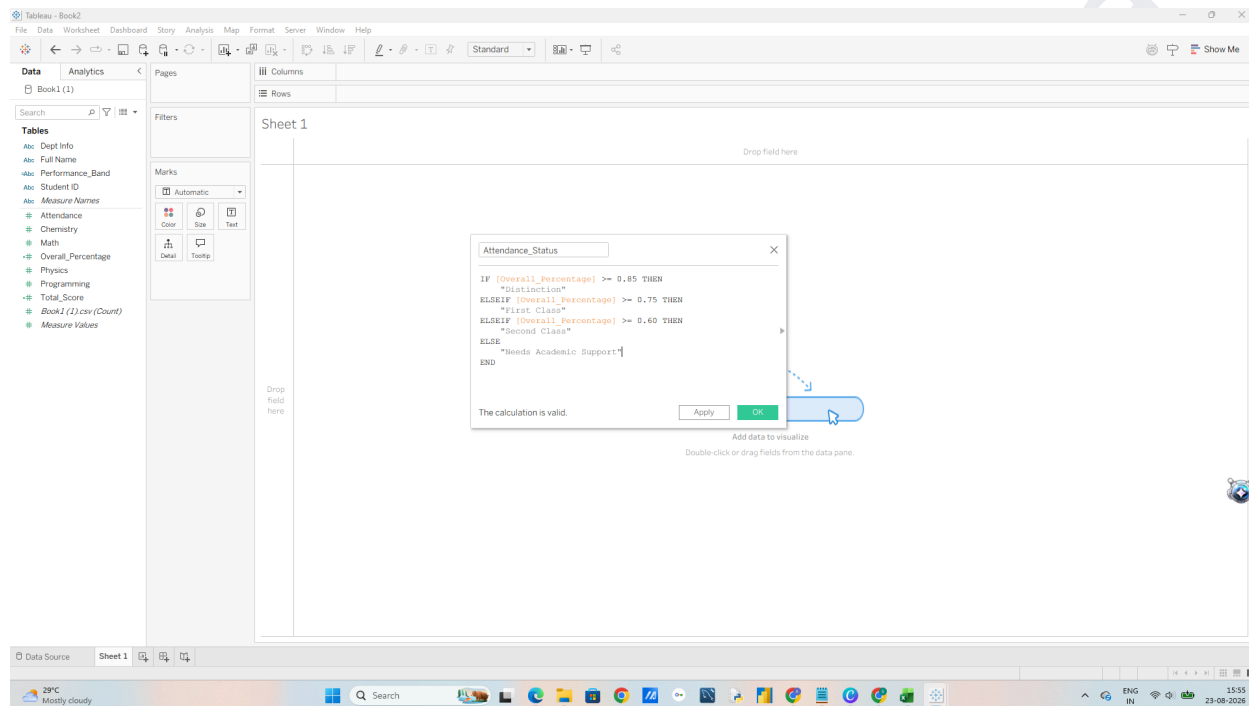
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Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane -->select Create Calculated Field....
2. Create the academic grading tier field:
 - o Field Name: Performance_Band
 - o Tableau Formula:



Step 3: Create an Attendance Health Flag 1. Right-click in the Data Pane -->select Create Calculated Field....

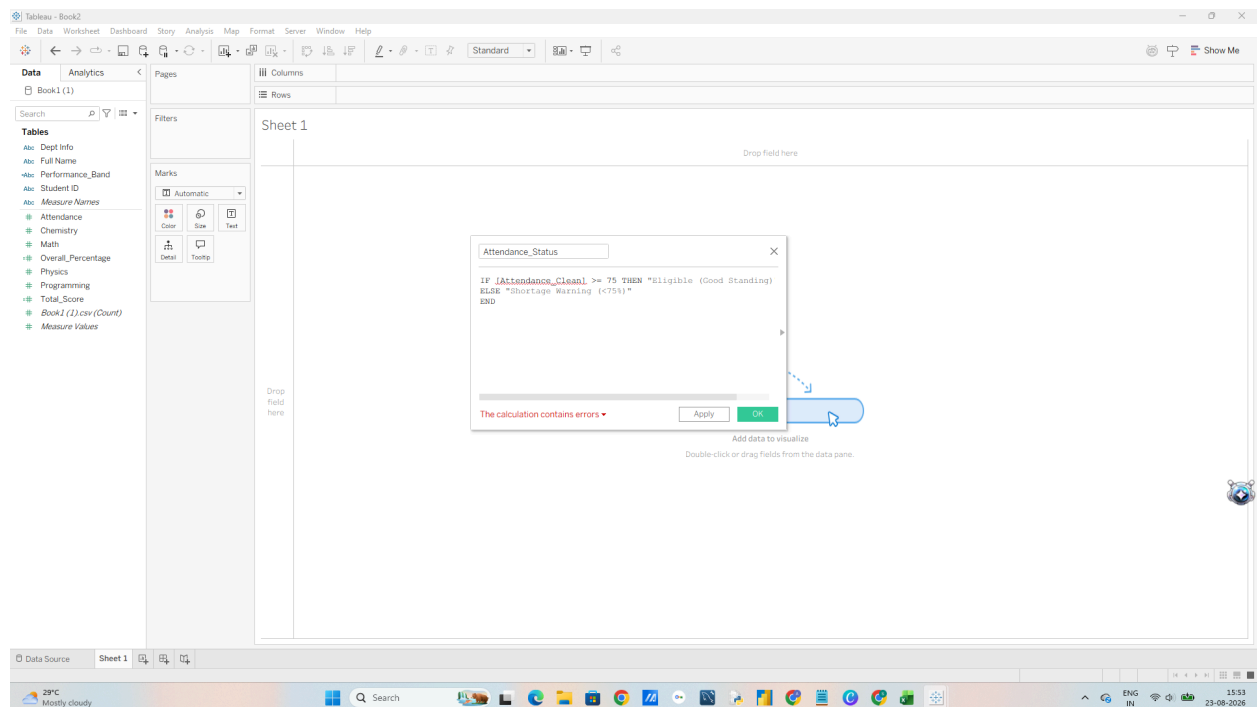
2. Create the attendance compliance indicator:
 - o Field Name: Attendance_Status
 - o Tableau Formula: Code snippet IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END
 - o Click O

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Step 4: Build the Student Performance Scorecard View

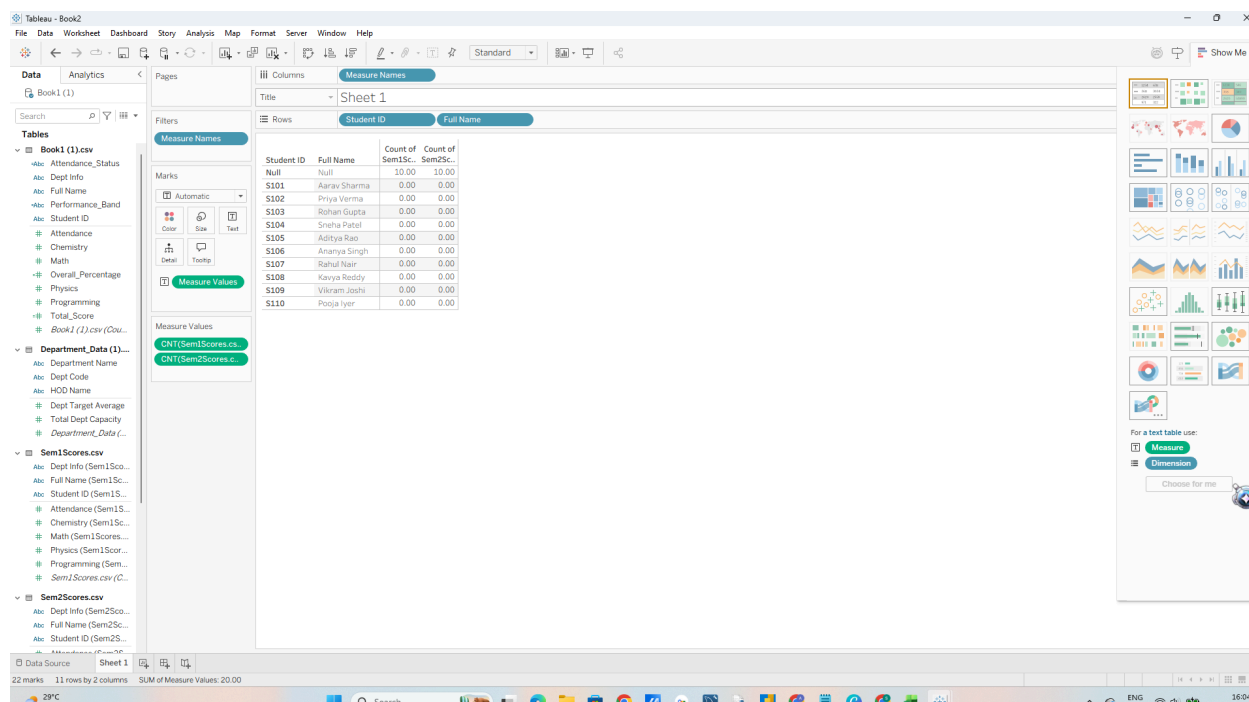
1. Drag Student_ID and Full_Name onto the Rows shelf.
2. Drag Semester onto the Columns shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.
4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card

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F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....

3. Create the department-wide benchmark field: o Field Name:

Dept_Average_Percentage_LOD

o Tableau Formula:

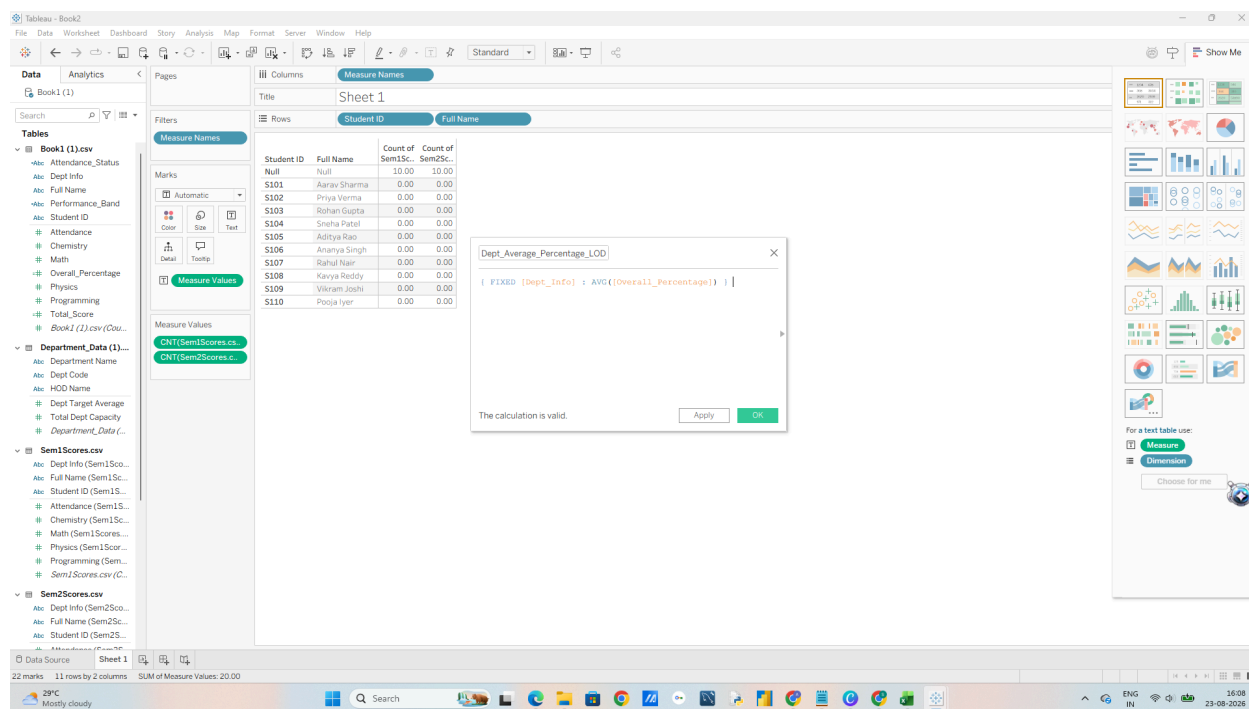
{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }

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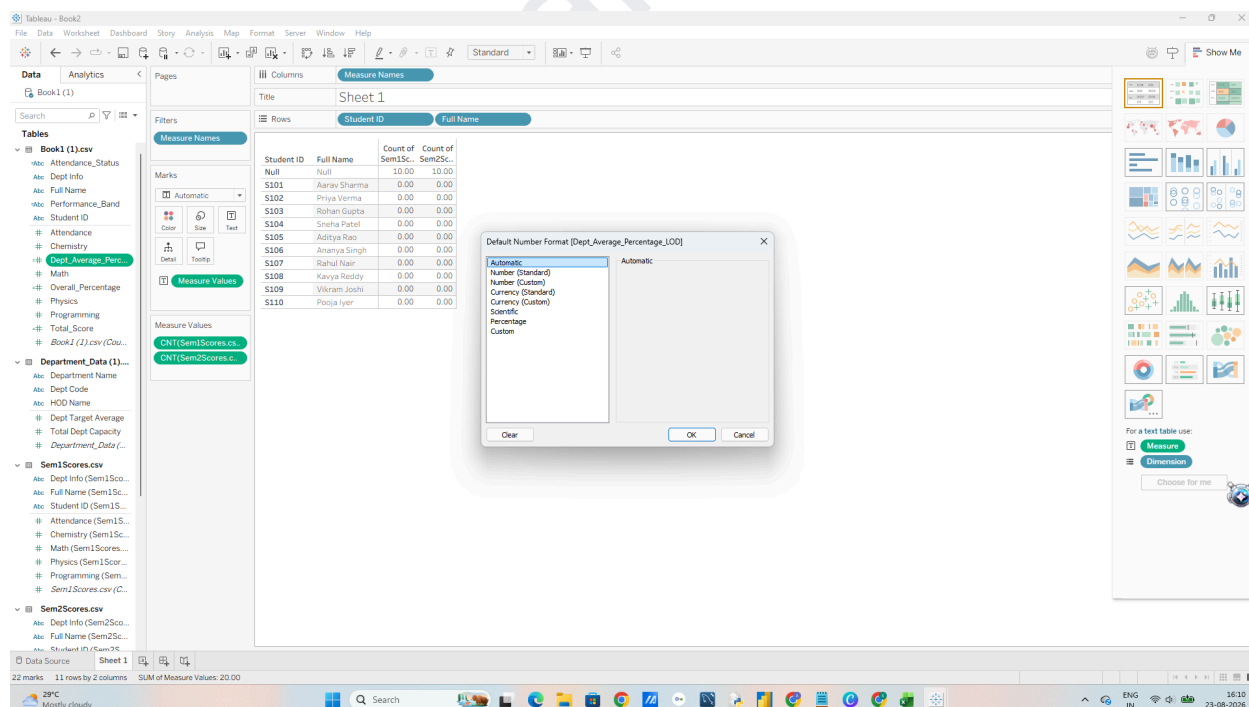
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4. Set default formatting: o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.



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Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

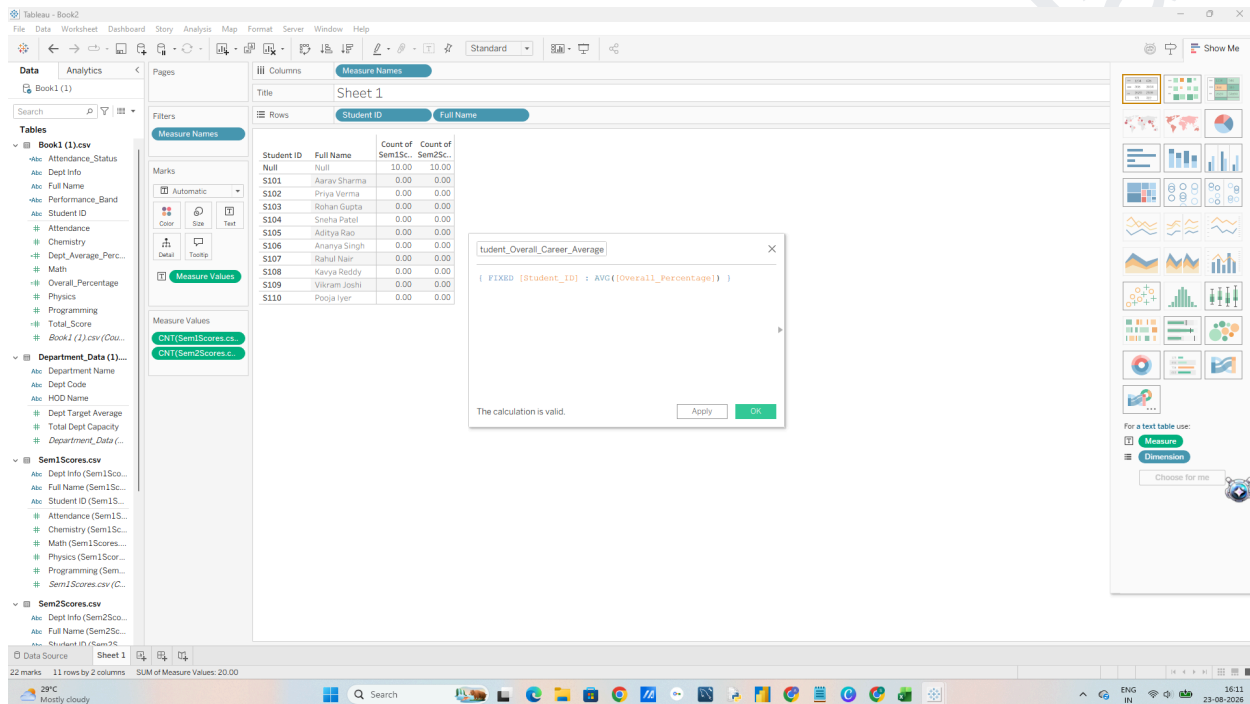
1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the cumulative student benchmark across all semesters: o Field Name:

Student_Overall_Career_Average

o Tableau Formula: Code snippet { FIXED [Student_ID] : AVG([Overall_Percentage]) }

o Click OK



Set its Number Format to Percentage (1 decimal place, Just like we did in previous step). Step

3: Calculate Performance Variance (Student vs. Department Benchmark) 1. Right-click in the

Data Pane --> select Create Calculated Field....

2. Create the relative difference metric: o Field Name: Variance_From_Dept_Average o Tableau

Formula: Code snippet [Overall_Percentage] - [Dept_Average_Percentage_LOD]

o Click OK.

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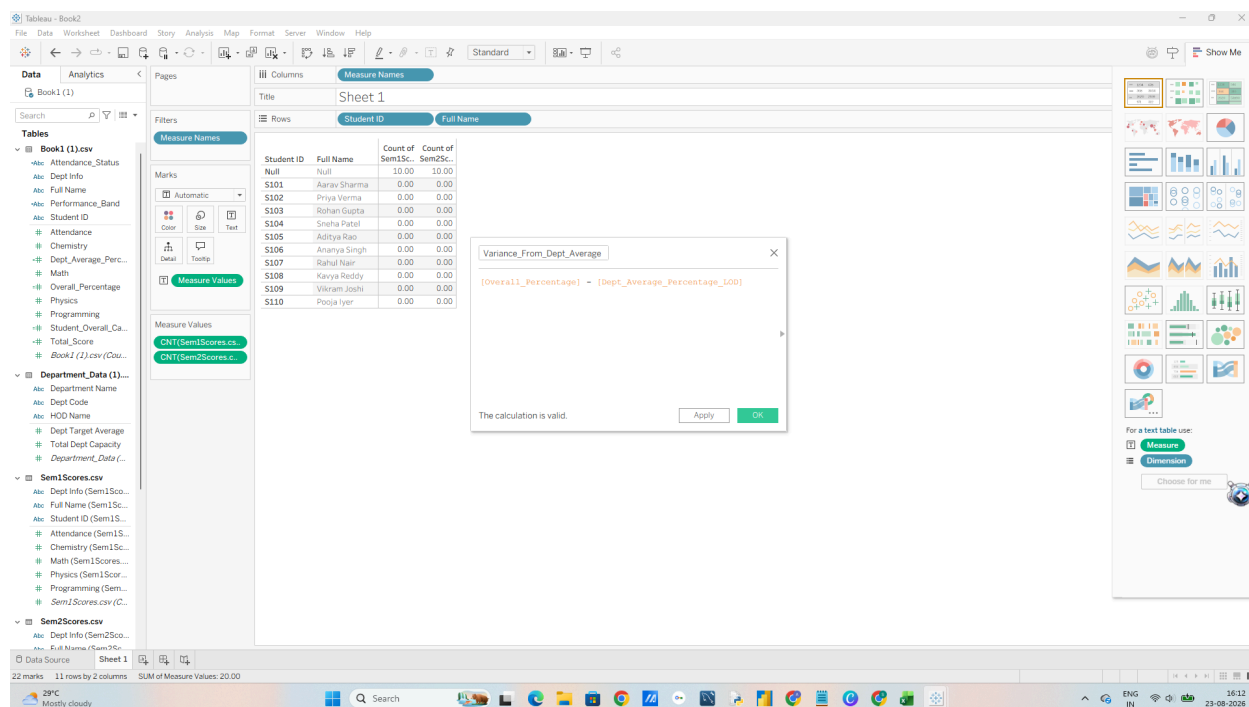
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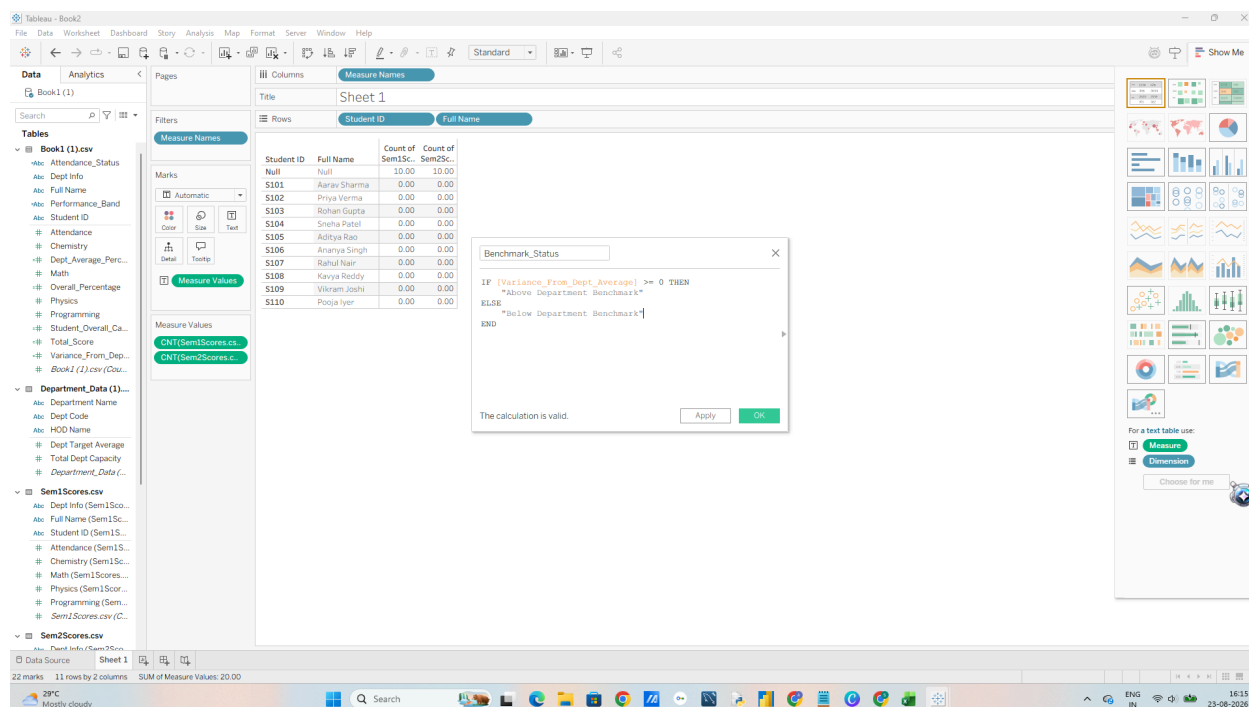
- Step 4: Create a Benchmark Indicator Dimension 1. Right-click in the Data Pane -->select Create Calculated Field
2. Create an automated comparison label: o Field Name: Benchmark_Status o Tableau Formula: Code snippet IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END o Click OK.

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Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.
2. Drag Semester to the Columns shelf.
3. Drag Overall_Percentage onto Text on the Marks card.
4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.
5. Drag Benchmark_Status onto Color on the Marks card.

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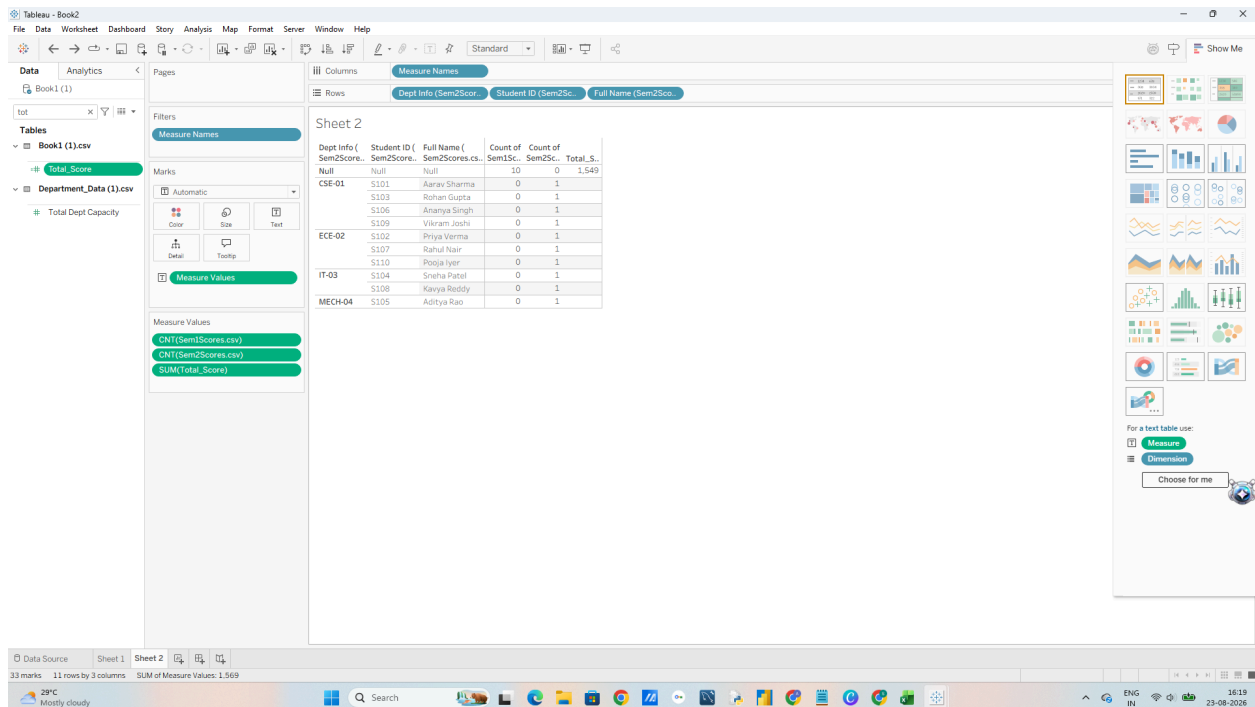
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G: Apply Advanced Data Cleaning Techniques

Objective: Parse, standardize, and extract structured components from irregular text fields (such as leading/trailing whitespace, inconsistent letter casing, and delimited composite strings). In student records, names often contain formatting defects (e.g., ' Aarav Sharma ' or 'priya VERMA'),

while operational codes like 'CSE-01' combine the academic branch and batch number in a single column. Applying text-parsing functions like TRIM(), SPLIT(), and casing standardizations ensures clean labels and enables multi-level categorization without altering the raw database files.

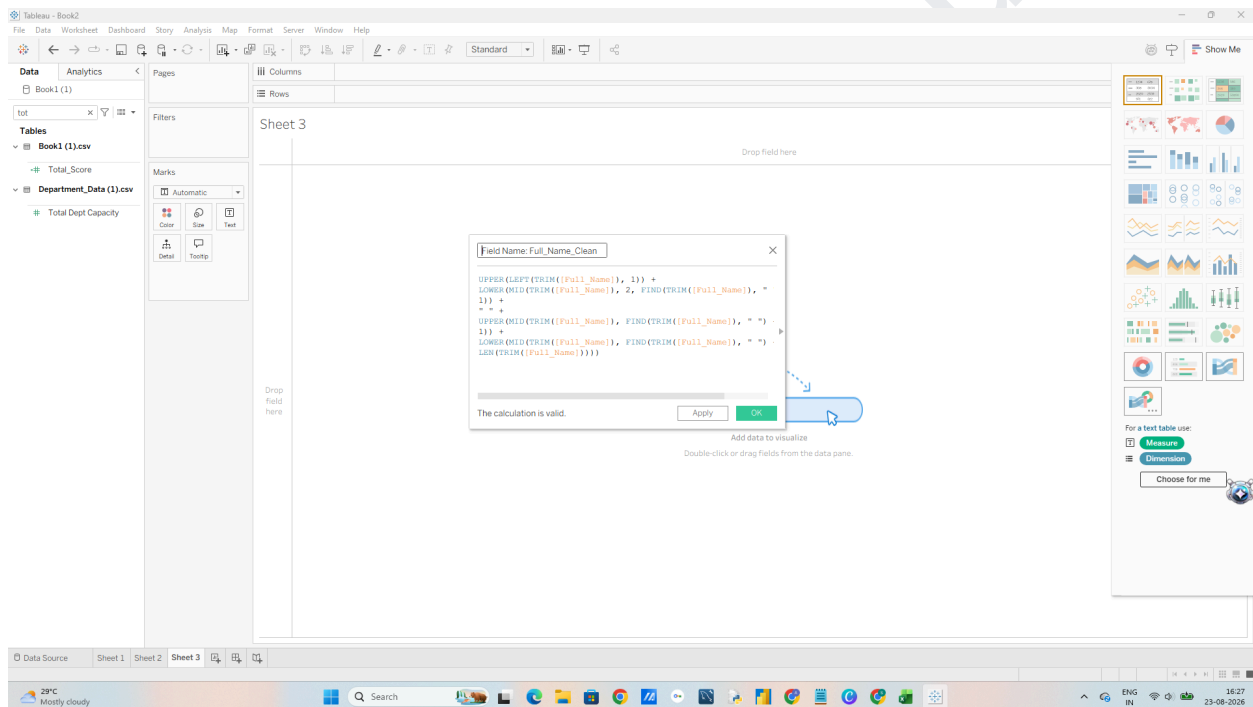
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Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

1. Navigate to Sheet 1 (or a new worksheet).
2. Right-click an empty space in the Data Pane --> select Create Calculated Field....
3. Create the trimmed and standardized student name field:
 - o Field Name: Full_Name_Clean
 - o Tableau Formula: Code snippet `UPPER(LEFT(TRIM([Full_Name]), 1)) + LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") - 1)) + " " + UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1, 1)) + LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2, LEN(TRIM([Full_Name]))))`
 - o (Alternatively, for standard trim-only cleaning): Code snippet `TRIM([Full_Name])`
 - o Click OK.



Step 2: Parse First Name and Last Name using SPLIT()

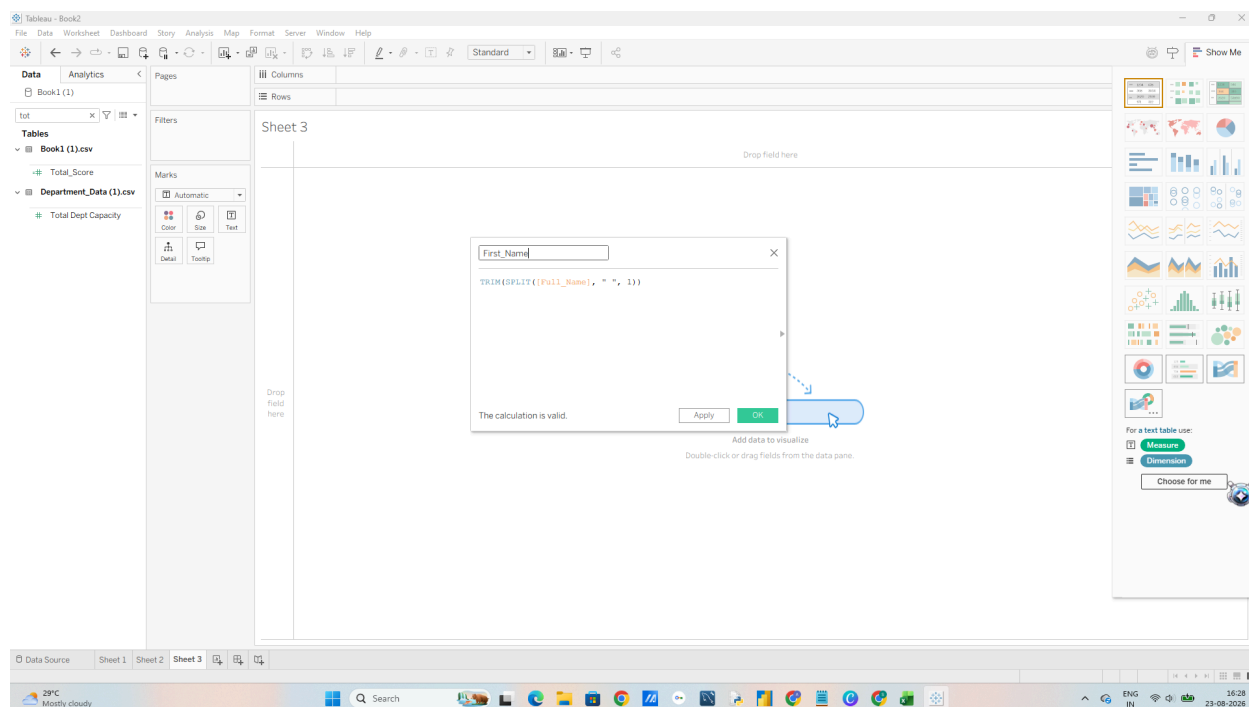
1. Extract the First Name:
 - o Right-click in the Data Pane --> select Create Calculated Field....
 - o Field Name: First_Name
 - o Tableau Formula: Code snippet `TRIM(SPLIT([Full_Name], " ", 1))`
 - o Click OK.

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Step 3: Split Composite Department Codes (Dept_Info) The column Dept_Info contains combined strings like CSE-01, ECE-02, and MECH-04. Separate them into distinct dimensional attributes: 1. Extract the Department Acronym:

- o Right-click in the Data Pane --> select Create Calculated Field....

- o Field Name: Dept_Branch o Tableau Formula: Code snippet SPLIT([Dept_Info], "-", 1)

- o Click OK.

2. Extract the Batch / Section ID:

- o Right-click in the Data Pane --> select Create Calculated Field....

- o Field Name: Section_ID

- o Tableau Formula: Code snippet SPLIT([Dept_Info], "-", 2)

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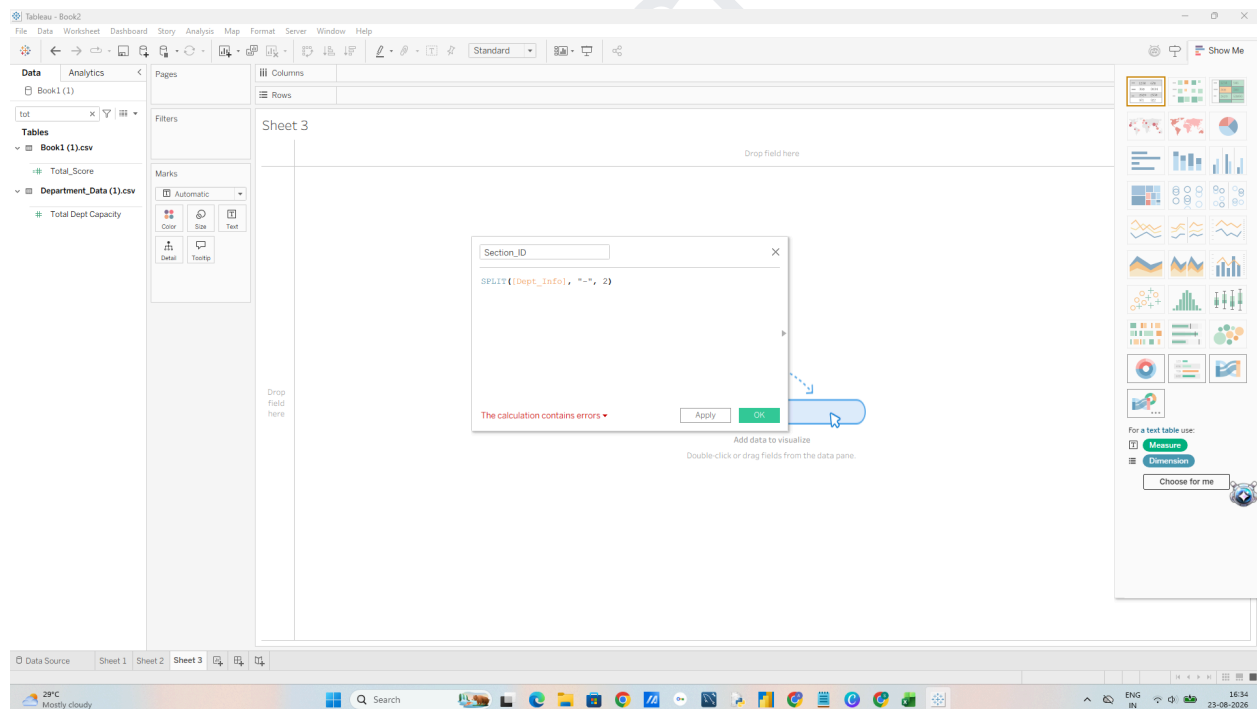
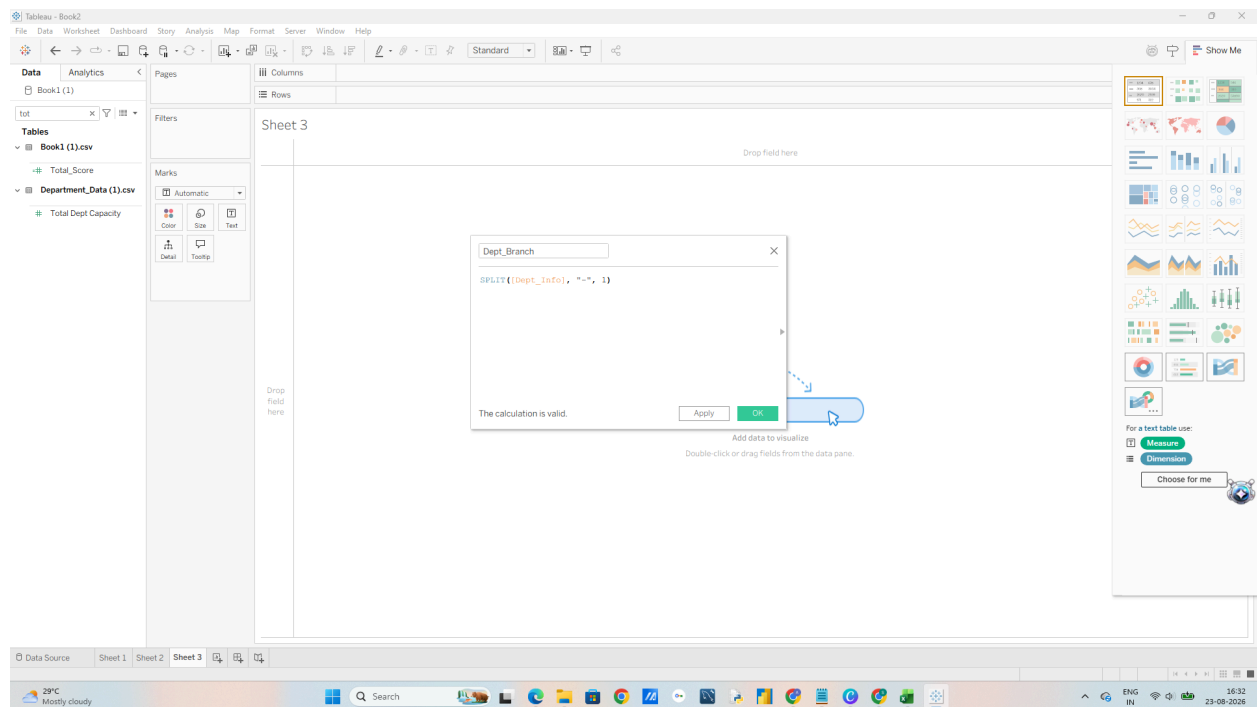
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Step 4: Create a Clean Composite Student Label Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Field Name: Student_Display_Label
3. Tableau Formula: Code snippet [Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"
4. Click OK

Step 5: Verify in the Final Dashboard View

1. Place Student_Display_Label on the Rows shelf.
2. Place Semester on the Columns shelf.
3. Place Overall_Percentage on Text.
4. Place Performance_Band on Color.
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)

Dept Info	Student ID	Full Name (Sem1Scores.csv)	Aarav Sharma	Aditya Rao	Ananya Singh	Kavya Reddy	Pooja Iyer	Priya Verma	Rahul Nair	Rohan Gupta	Sneha Patel
CSE-01	S101	Aarav Sharma	1.000	1.000	1.000	1.000				1.000	1.000
		Ananya Singh	1.000	1.000	1.000	1.000				1.000	1.000
		Rohan Gupta	1.000	1.000	1.000	1.000				1.000	1.000
		Vikram Joshi	1.000	1.000	1.000	1.000				1.000	1.000
	S103	Aarav Sharma	1.000	1.000	1.000	1.000				1.000	1.000
		Ananya Singh	1.000	1.000	1.000	1.000				1.000	1.000
		Rohan Gupta	1.000	1.000	1.000	1.000				1.000	1.000
		Vikram Joshi	1.000	1.000	1.000	1.000				1.000	1.000
	S106	Aarav Sharma	1.000	1.000	1.000	1.000				1.000	1.000
		Ananya Singh	1.000	1.000	1.000	1.000				1.000	1.000
		Rohan Gupta	1.000	1.000	1.000	1.000				1.000	1.000
		Vikram Joshi	1.000	1.000	1.000	1.000				1.000	1.000
	S109	Aarav Sharma	1.000	1.000	1.000	1.000				1.000	1.000
		Ananya Singh	1.000	1.000	1.000	1.000				1.000	1.000
		Rohan Gupta	1.000	1.000	1.000	1.000				1.000	1.000
		Vikram Joshi	1.000	1.000	1.000	1.000				1.000	1.000
ECE-02	S102	Pooja Iyer					1.000	1.000	1.000	1.000	1.000
		Priya Verma					1.000	1.000	1.000	1.000	1.000
		Rahul Nair					1.000	1.000	1.000	1.000	1.000
	S107	Pooja Iyer					1.000	1.000	1.000	1.000	1.000
		Priya Verma					1.000	1.000	1.000	1.000	1.000
		Rahul Nair					1.000	1.000	1.000	1.000	1.000
	S110	Pooja Iyer					1.000	1.000	1.000	1.000	1.000
		Priya Verma					1.000	1.000	1.000	1.000	1.000
		Rahul Nair					1.000	1.000	1.000	1.000	1.000
IT-03	S104	Kavya Reddy				1.000	1.000				1.000
		Sneha Patel				1.000	1.000				1.000
	S108	Kavya Reddy				1.000	1.000				1.000
		Sneha Patel				1.000	1.000				1.000
MECH-04	S105	Aditya Rao		1.000	1.000						1.000

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